Pokhara University

Faculty of science and technology

Subject . Digital Logic Capsule

Computer Engineering I Semester

Software Engineering I Semester

Ele and Eletrical Engineering III Semester

Name..

Subject..

Semester..

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By. Er. Manoj Kumar Singh

Availeval. Bijayapur library

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Design a combinational CKT which takes three i/p numbers and produces an output equal to square of the input. [2013 Fall]

Ans: Analysis.

Max No. = 111 = 7 square of $7 = 7^2 = 49$

49 in Binary = 110001

Total no. of bits in O/P seq = 6.

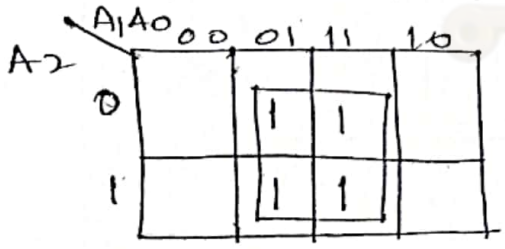
	I/P			O/P					
	A_2	A_1	A_0	S_5	S_4	S_3	S_2	S_1	S_0
0	0	0	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0	0	1
2	0	1	0	0	0	0	1	0	0
3	0	1	1	0	0	1	0	0	1
4	1	0	0	0	1	0	0	0	0
5	1	0	1	0	1	1	0	0	1
6	1	1	0	1	0	0	1	0	0
7	1	1	1	1	1	0	0	0	1

square of 0 = 0
square of 1 = 1
square of 2 = 4
square of 3 = 9
square of 4 = 16
square of 5 = 25
square of 6 = 36
square of 7 = 49

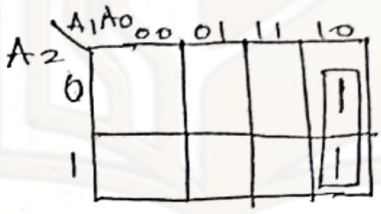
Truth table

[S_1 = 0 so need to make K-Map]

Now let us draw the K-Map for S_0 for S_2



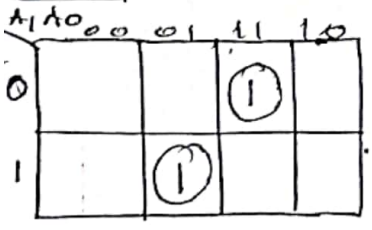
$S_0 = A_0$



$S_2 = A_1\bar{A}_0$

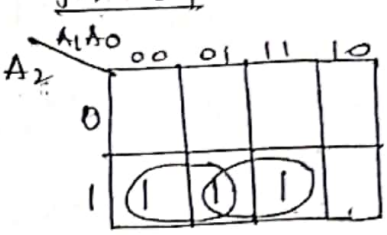
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for S_3

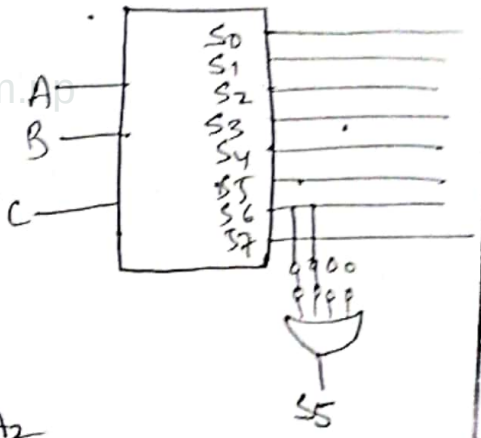


$S_3 = A_0A_1\bar{A}_2 + A_0\bar{A}_1A_2$

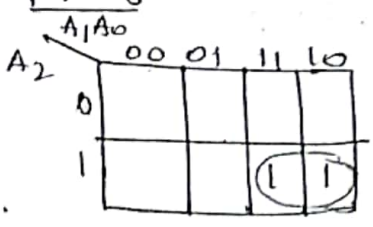
for S_4



$S_4 = \bar{A}_1A_2 + A_0A_2$



for S_5



$S_5 = A_1A_2$

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Now, let us draw the logic diagram

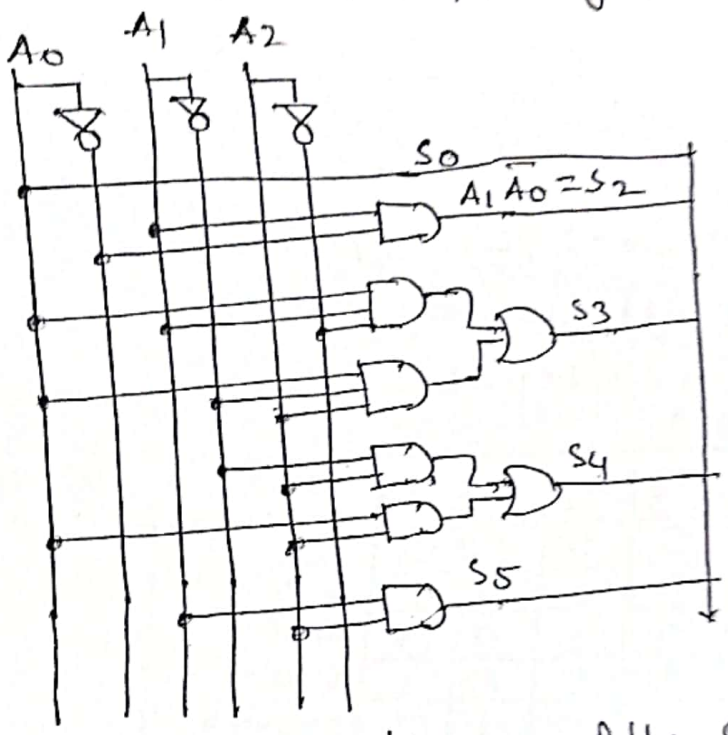


Fig:- logic diagram of the ~~gikon ckt~~ 3 bit square o/p.

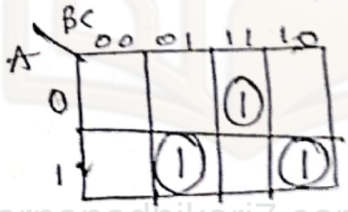
2. Design a combinational logic ckt with 3 inputs that pro 1 o/p when exactly two variables are 1.

Sol:- Let us draw the Truth Table

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Truth Table.

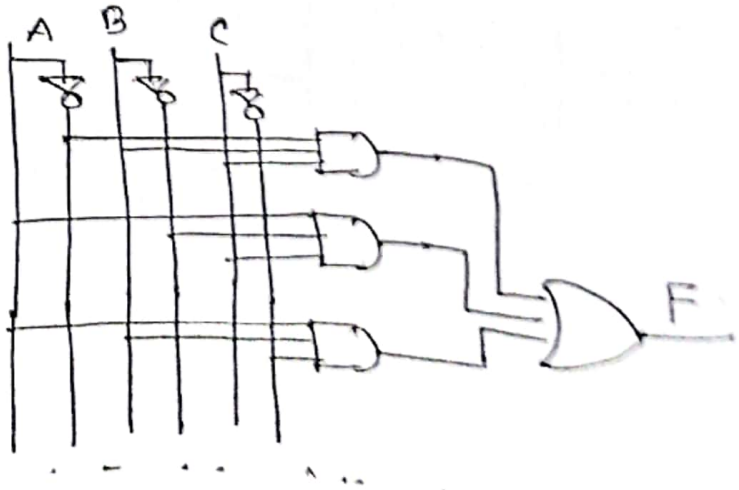
Now K-Map for F



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$F = \bar{A}BC + A\bar{B}C + AB\bar{C}$

Now draw the logic diagram.



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Q. Design a combinational ckt with three inputs x, y, z and three outputs A, B, C when the binary input is 0, 1, 2 or 3, the binary output is two greater than the input. When the binary input is 4, 5, 6 or 7, the binary output is two less than the input.

Sol^n →

	x	y	z	A	B	C
0	0	0	0	0	1	0
1	0	0	1	0	1	1
2	0	1	0	1	0	0
3	0	1	1	1	0	1
4	1	0	0	0	1	0
5	1	0	1	0	1	1
6	1	1	0	1	0	0
7	1	1	1	1	0	1

Truth Table.

Note
ACC. to given
0, 1, 2 or 3
Add 2 to
4, 5 6 or 7
subtract 2
i.e. 4-2, 5-2
6-2

Now, K Map for Output A

x \ yz	00	01	11	10
0			1	1
1			1	1

A = y

K-Map for Output B

x \ yz	00	01	11	10
0	1	1		
1	1	1		

B = y'

Again K-Map for Output C

x \ yz	00	01	11	10
0		1	1	
1		1	1	

C = z

Now draw the logic diagram

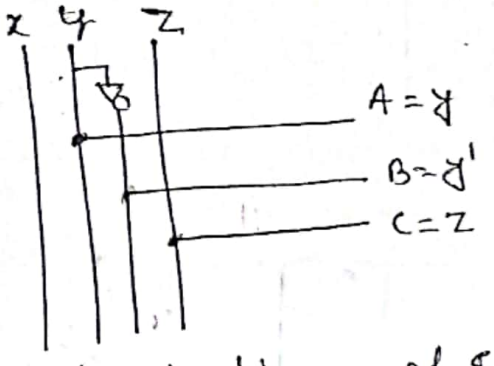


Fig. logic diagram of given circuit

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Design a combinational ckt with three inputs and one output.

(a) The output is 1 when binary value of the inputs is less than or equal to 3. The output is 0 otherwise.

(b) The output is 1 when the binary value of the inputs is an odd number.

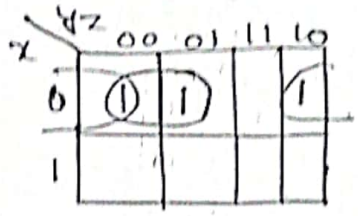
(c) The output is 1 when the binary value of the inputs is an even numbers.

Solution :→ (a)

x	y	z	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

Truth table.

Now, write the K-Map for F



$F = x'y' + x'z'$.

Now, Draw the logic diagram of given Boolean function.

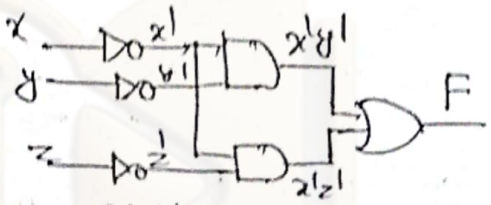
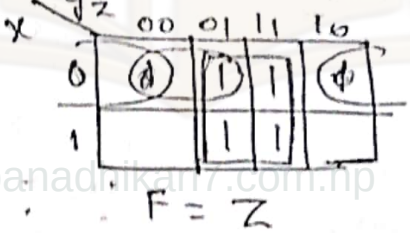


Fig: logic diagram

(b) for odd No. o/p = 1

	x	y	z	F
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	1
4	1	0	0	0
5	1	0	1	1
6	1	1	0	0
7	1	1	1	1

K-Map for o/p F



$F = z$

Now logic ckt

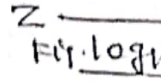
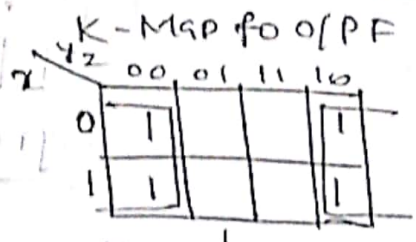


Fig: logic ckt

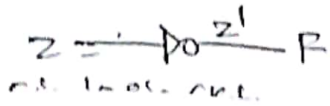
(c) for even No. o/p = 1

	x	y	z	F
0	0	0	0	1
1	0	0	1	0
2	0	1	0	1
3	0	1	1	0
4	1	0	0	1
5	1	0	1	0
6	1	1	0	1
7	1	1	1	0

K-Map for o/p F



$F = z'$



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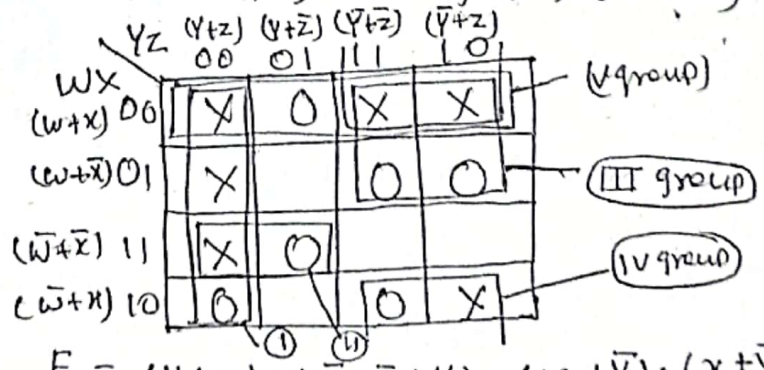
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$F(w,x,y,z) = \sum(1,6,7,8,11,13)$ with don't care condition
$d(w,x,y,z) = \sum(0,2,3,4,10,12)$ then implement the functio
using NOR gates only. [2012 Fall]

Sol: Here given:

$$F(w,x,y,z) = \sum(1,6,7,8,11,13)$$
$$d(w,x,y,z) = \sum(0,2,3,4,10,12)$$



$$F = (y+z)\cdot(\bar{w}+\bar{x}+y)\cdot(w+\bar{y})\cdot(x+\bar{y})\cdot(w+x)$$

Now, Double complementing on Both side we set

$$\bar{\bar{F}} = \overline{(y+z)\cdot(\bar{w}+\bar{x}+y)\cdot(w+\bar{y})\cdot(x+\bar{y})\cdot(w+x)}$$
$$= \overline{(\bar{y}+z)} + \overline{(\bar{w}+\bar{x}+y)} + \overline{(w+\bar{y})} + \overline{(x+\bar{y})} + \overline{(\bar{w}+x)}$$

Now, Implementing the function using NOR gate only.

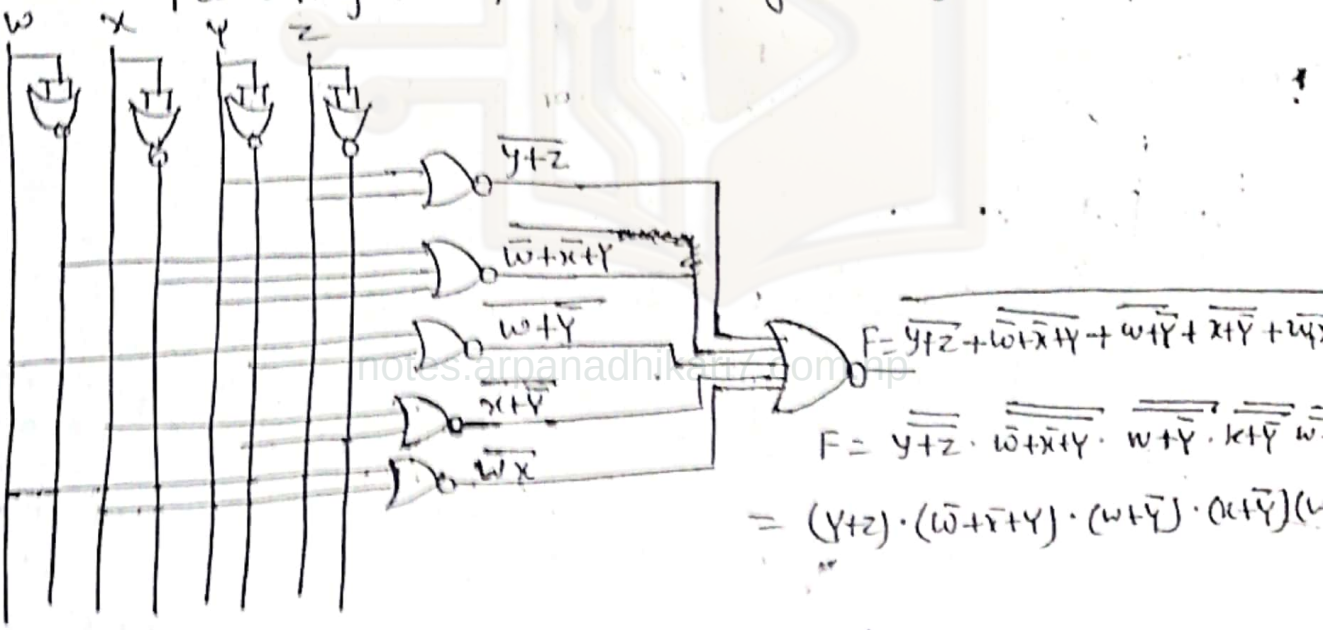


Fig: logic diagram with NOR gate only.

Prepared by: D Manoj for Sirjal.

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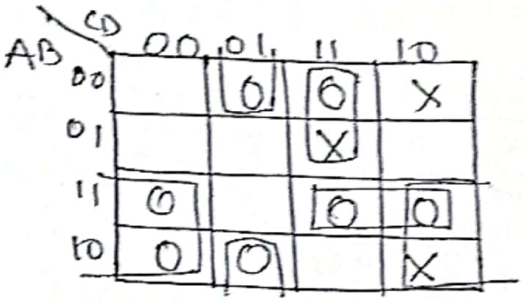
* Use K-Map to simplify the given [illegible] function in POS & implement the simplified function using NOR gate only.

$F(A,B,C,D) = \sum(1,3,8,9,12,13,14,15)$ & don't care $d(A,B,C,D) = \sum(2,7,10)$ (2012 Spring)

Sol:⇒ Here, Given

$F(A,B,C,D) = \sum(1,3,8,9,12,13,14,15)$

$d(A,B,C,D) = \sum(2,7,10)$



$F = (B+C+\bar{D}) \cdot (A+\bar{C}+\bar{D}) \cdot (\bar{A}+\bar{B}+\bar{C}) \cdot (\bar{A}+D)$

Now, ~~Implementing this function by using NOR gate only~~

Now, Double complementing On Both side we get

$$\bar{\bar{F}} = \overline{(B+C+\bar{D}) \cdot (A+\bar{C}+\bar{D}) \cdot (\bar{A}+\bar{B}+\bar{C}) \cdot (\bar{A}+D)}$$

$$= \overline{(\overline{B+C+\bar{D}}) + (\overline{A+\bar{C}+\bar{D}}) + (\overline{\bar{A}+\bar{B}+\bar{C}}) + (\overline{\bar{A}+D})}$$

Now, Implementing this function by using NOR gate only

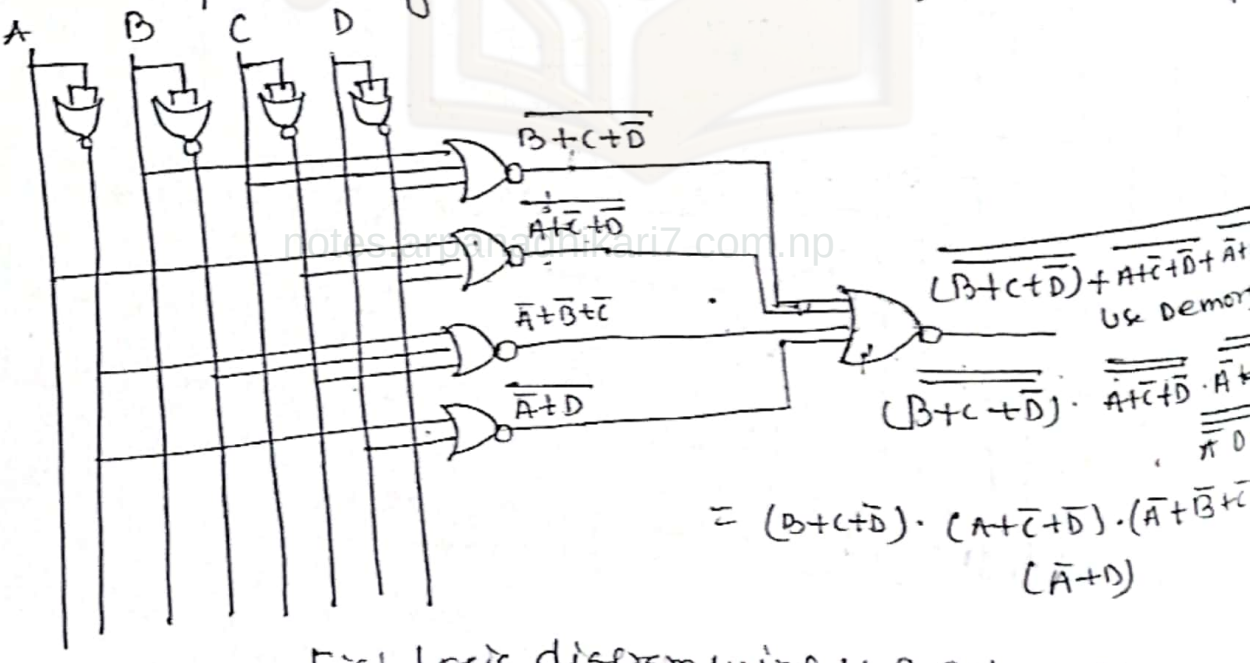


Fig: Logic diagram using NOR gates only.

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(1) Design a logic function ckt to implement th Boolean function. 2

$F(A,B,C,D) = \sum(1,3,7,11,15)$

$D(A,B,C,D) = \sum(0,2,5)$ in the term of

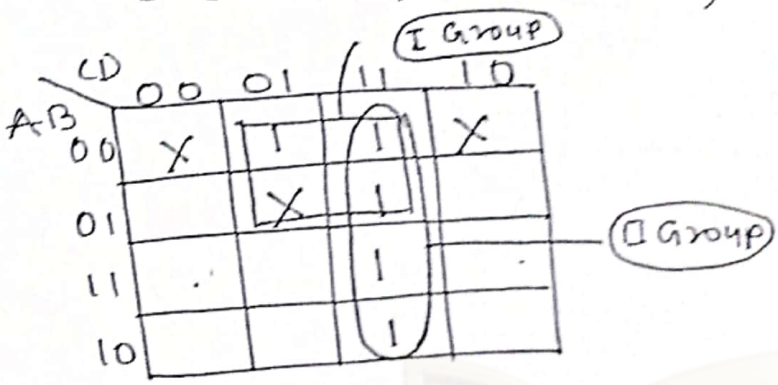
(i) SOP

(ii) implement with NAND-NAND gate only. [2013 Fall]

Soln:⇒

$F(A,B,C,D) = \sum(1,3,7,11,15)$

$D(A,B,C,D) = \sum(0,2,5)$



$F = \bar{A}D + C\bar{D}$

Now, Double complementing on Both side we get

$\bar{\bar{F}} = \overline{\overline{\bar{A}D + C\bar{D}}}$

$= \overline{\overline{\bar{A}D} \cdot \overline{C\bar{D}}}$ (using De morgan's theorem)

Now, Implement by using NAND gate only.

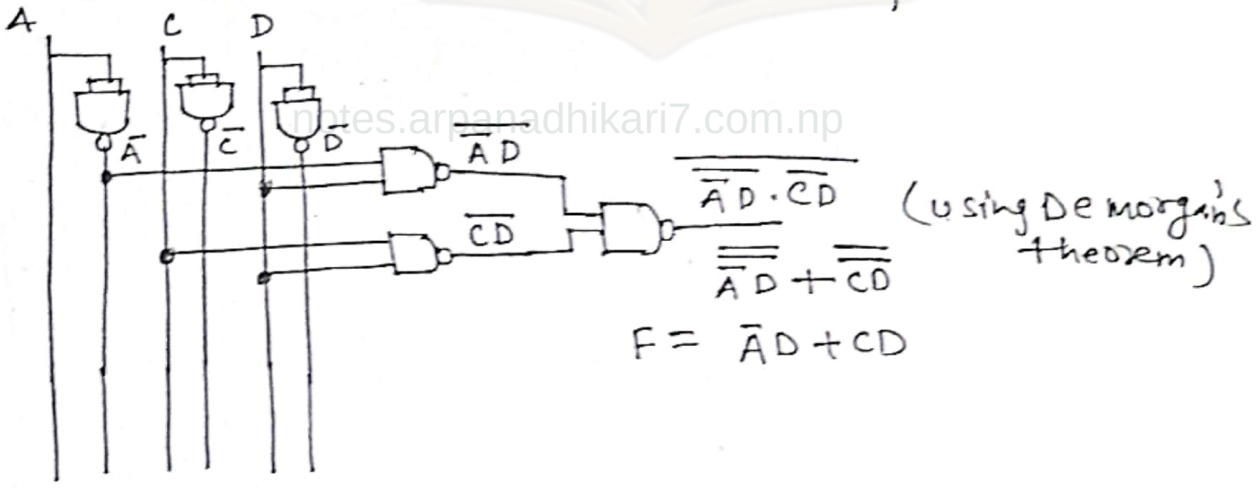


Fig:- logic diagram using NAND gate only

Prepared By:- ER. Manoj Kumar Singh

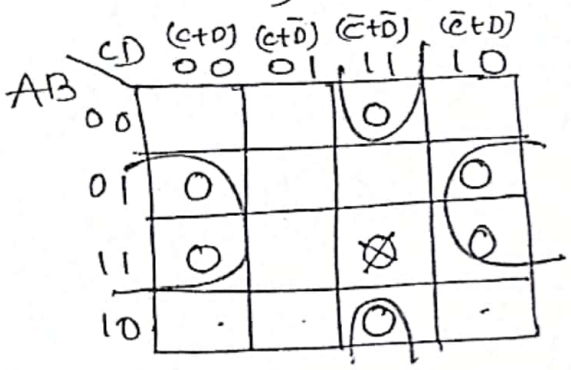
2

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(2) Obtain a reduced expression for the following $F(A,B,C,D) = \sum m(0,1,2,5,7,8,9,10,13,15)$ and realize the minimized function using NOR gate only.

Soln → Here,

$$F(A,B,C,D) = \sum m(0,1,2,5,7,8,9,10,13,15)$$



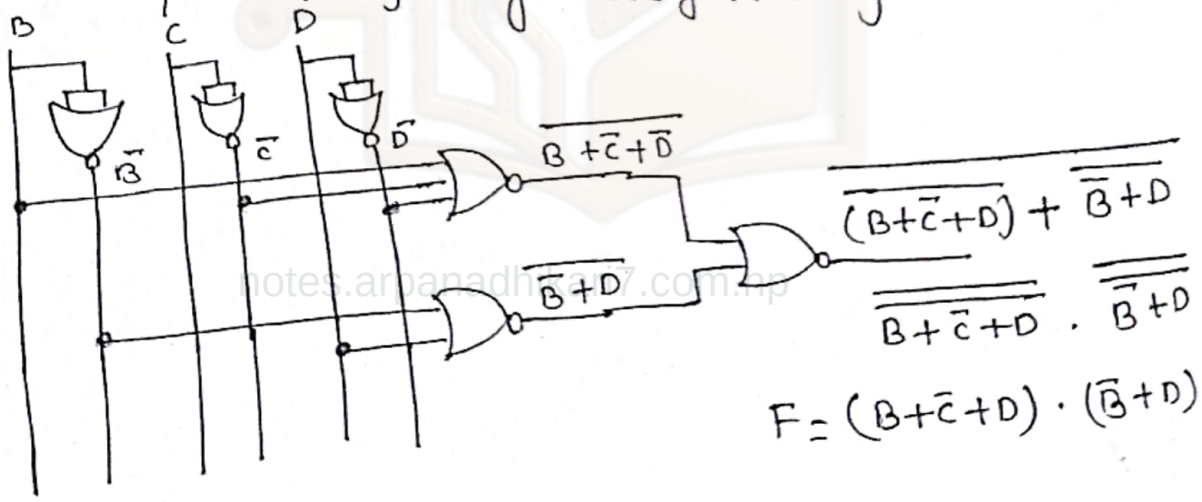
$$F = (B+\bar{C}+\bar{D})\cdot(\bar{B}+D)$$

Now, Double complementing on both side we get

$$\bar{\bar{F}} = \overline{\overline{(B+\bar{C}+\bar{D})\cdot(\bar{B}+D)}}$$

$$= \overline{\overline{(B+\bar{C}+\bar{D})} + \overline{(\bar{B}+D)}}$$

Now, Implement by using NOR gate only.



$$F = (B+\bar{C}+D)\cdot(\bar{B}+D)$$

Fig:- logic diagram

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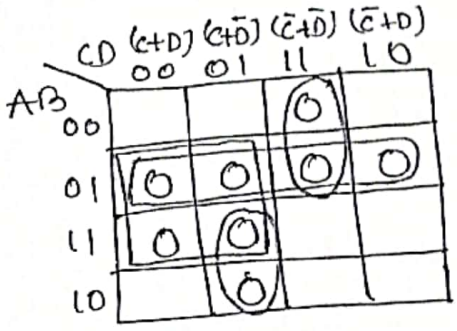
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(9) Obtain a ... for the following
$F(A,B,C,D) = \sum m(0,1,2,8,10,11,14,15)$ and realize the minimized function using NOR gate only. 3

⇒ Here,

$F(A,B,C,D) = \sum m(0,1,2,8,10,11,14,15)$ for Maxterm

$F(A,B,C,D) = \Pi M(3,4,5,6,7,9,13)$



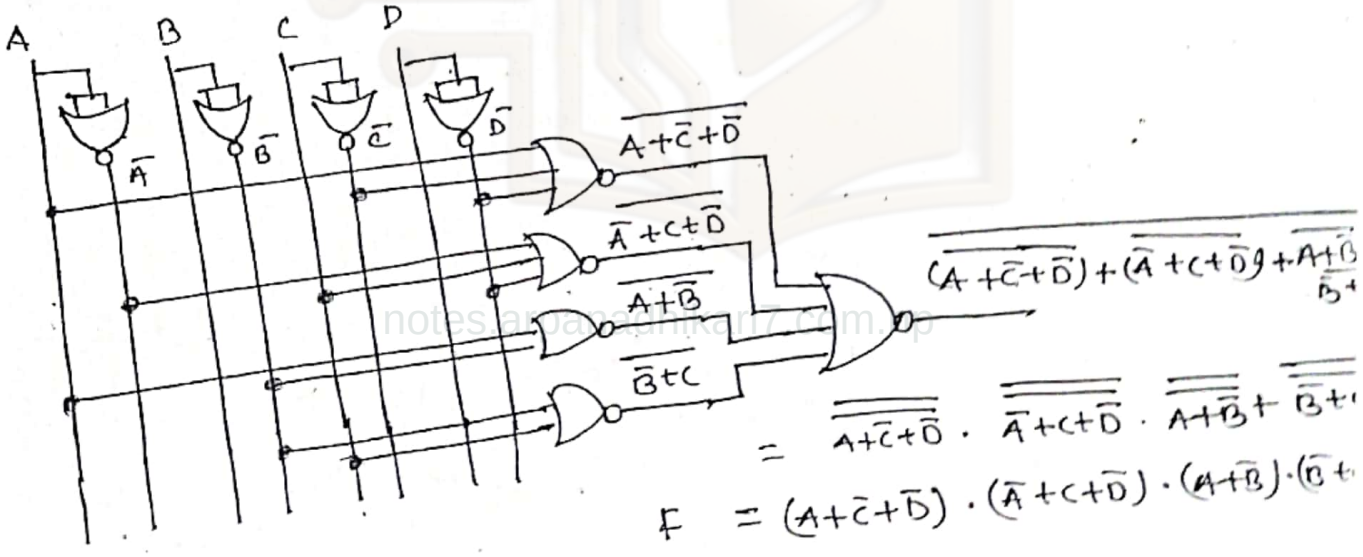
$F = (A+\bar{C}+\bar{D})\,(\bar{A}+C+\bar{D})\cdot(A+\bar{B})\cdot(\bar{B}+C)$

Now, Double complementing on Both Sides we get.

$$\bar{\bar{F}} = \overline{(A+\bar{C}+\bar{D})\cdot(\bar{A}+C+\bar{D})\cdot(A+\bar{B})\cdot(\bar{B}+C)}$$

$$= \overline{\overline{(A+\bar{C}+\bar{D})} + \overline{(\bar{A}+C+\bar{D})} + \overline{(A+\bar{B})} + \overline{(\bar{B}+C)}}$$

Now, Implement by using NOR gate only.



$$= \overline{\overline{A+\bar{C}+\bar{D}}} \cdot \overline{\overline{\bar{A}+C+\bar{D}}} \cdot \overline{\overline{A+\bar{B}}} + \overline{\overline{\bar{B}+C}}$$

$$F = (A+\bar{C}+\bar{D})\cdot(\bar{A}+C+\bar{D})\cdot(A+\bar{B})\cdot(\bar{B}+C)$$

Fig. logic diagram.

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(4) Design a logic circuit to implement the Boolean function.

$F(A,B,C,D) = \Sigma(1,3,4,5,7,9,13,14,15)$

$D(A,B,C,D) = \Sigma(0,2,8)$

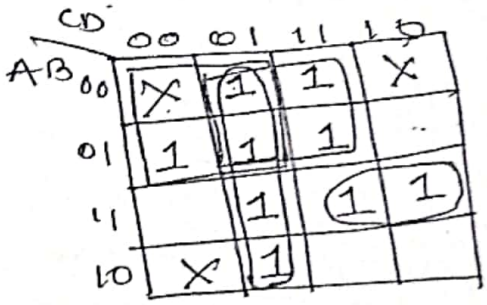
(i) Sum of product

(ii) Implement with NAND–NAND gate only.

Sol:→ Here, Given

$F(A,B,C,D) = \Sigma(1,3,4,5,7,9,13,14,15)$

$D(A,B,C,D) = \Sigma(0,2,8)$



$F = ABC + \bar{A}\bar{C} + \bar{A}D + \bar{C}D$

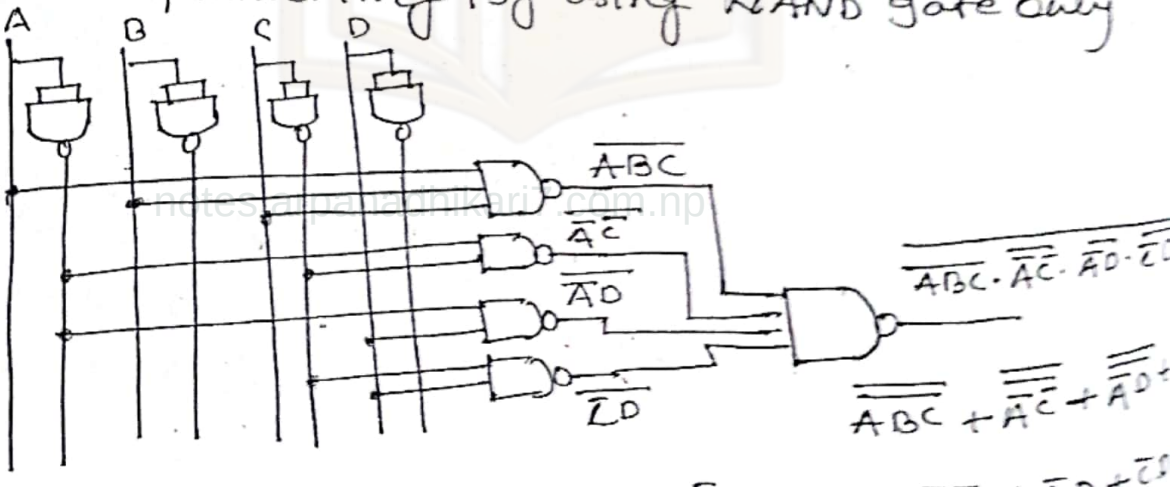
(i) SOP $F = ABC + \bar{A}\bar{C} + \bar{A}D + \bar{C}D$

Now, Double complementing on Both side we get

$$\overline{\overline{F}} = \overline{ABC + \bar{A}\bar{C} + \bar{A}D + \bar{C}D}$$

$$= \overline{\overline{ABC} \cdot \overline{\bar{A}\bar{C}} + \overline{\bar{A}D} + \overline{\bar{C}D}}$$

Now, Implementing By using NAND gate only



$F = ABC + \bar{A}\bar{C} + \bar{A}D + \bar{C}D$

Fig:- logic diagram.

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Q. A Boolean Function is given by $F(A,B,C,D) = \Sigma(3,4,5,7,9,13,14,15)$ and don't care condition $d(A,B,C,D) = \Sigma(0,2,8)$. Simplify it using K-Map and implement using NAND gate only. (2014 Fall) (4)

=) Here, (A,B,C,D)

Given $F = \Sigma(3,4,5,7,9,13,14,15)$

$d(A,B,C,D) = \Sigma(0,2,8)$

AB \ CD	00	01	11	10
00			1	
01	1	1	1	
11		1	1	1
10		1		

$$F = \bar{A}B\bar{C} + \bar{A}CD + A\bar{C}D + ABC$$

Now, Double complementing on Both side we get

$$\bar{\bar{F}} = \overline{\bar{A}B\bar{C} + \bar{A}CD + A\bar{C}D + ABC}$$

$$= \overline{\bar{A}B\bar{C} \cdot \bar{A}CD \cdot A\bar{C}D \cdot ABC}$$

Now, Implement by using NAND gate only.

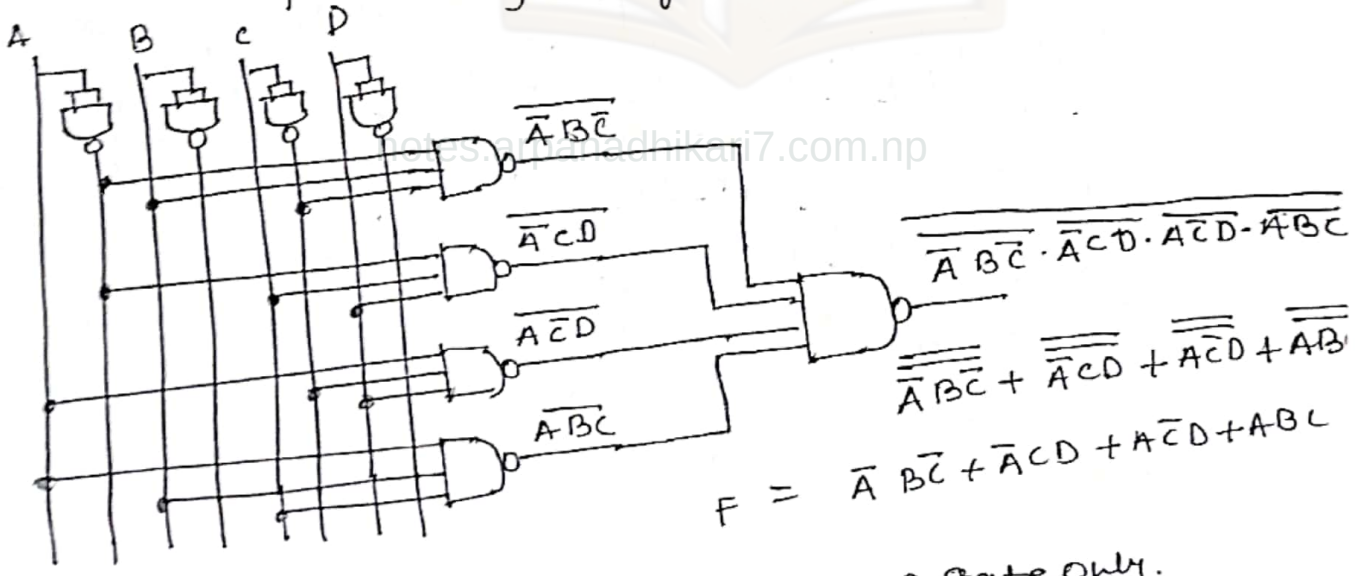


Fig: logic diagram using NAND gate only.

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* A Boolean function is given by $F(A,B,C,D) = \Sigma(3,4,6,8,10,12,14)$ and don't care condition $d(A,B,C,D) = \Sigma(0,2,7)$. Simplify it using K-Map & implement using NAND gate only.

[2015 Fall / Spring]

Sol:⇒ Here, Given

$F(A,B,C,D) = \Sigma(3,4,6,8,10,12,14)$

$d(A,B,C,D) = \Sigma(0,2,7)$

AB \ CD	00	01	11	10
00	X		1	X
01	1		X	1
11	1			1
10	1			1

$F = \bar{A}C + \bar{D}$

Now, Double complementing on Both side we get

$\overline{\overline{F}} = \overline{\bar{A}C + \bar{D}}$

$= \overline{\overline{\bar{A}C} \cdot \overline{\bar{D}}}$

Now Implement By using NAND gate only.

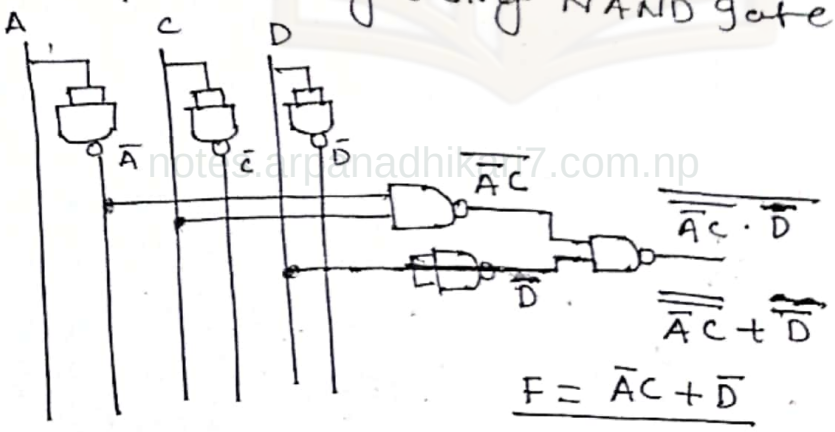


Fig:- Logic diagram

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Q. Simplify given function using K-Map & implement it in a circuit. 5

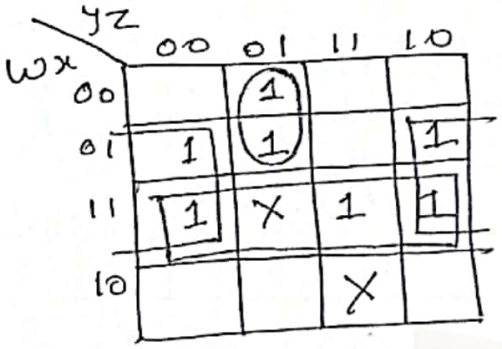
$F(W,X,Y,Z) = \Sigma(1,4,5,6,12,14,15)$ & don't care condition $D(W,X,Y,Z) = \Sigma(11,13)$

[2016 spring]

Sol^n ⇒

$F(W,X,Y,Z) = \Sigma(1,4,5,6,12,14,15)$

$D(W,X,Y,Z) = \Sigma(11,13)$



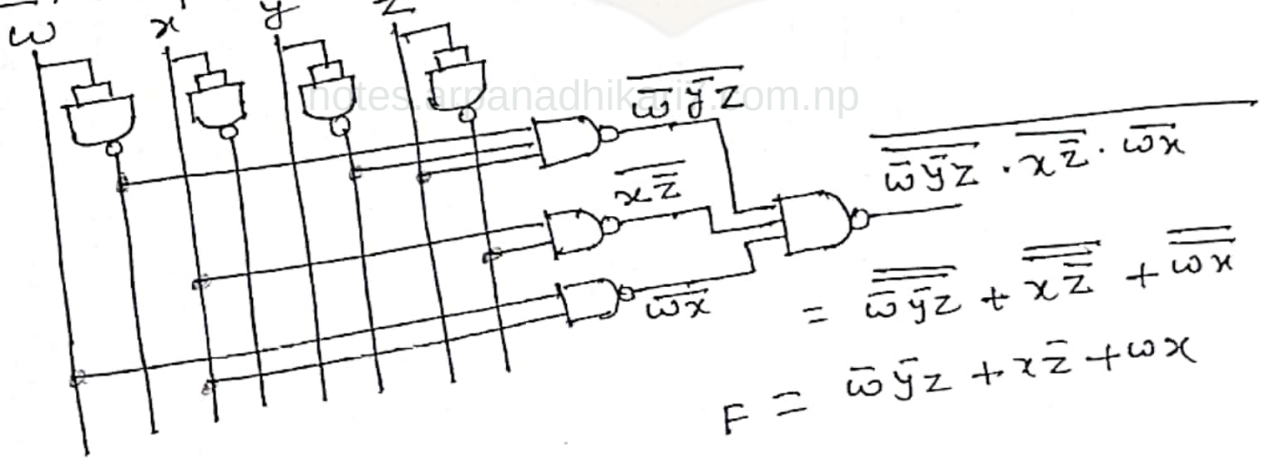
$F = \bar{w}\bar{y}z + x\bar{z} + wx$

Now, Double complementing on both side we get

$$\bar{\bar{F}} = \overline{\overline{\bar{w}\bar{y}z + x\bar{z} + wx}}$$

$$= \overline{\bar{w}\bar{y}z \cdot \overline{x\bar{z}} \cdot \overline{wx}}$$

Now, Implement by using NAND gate



$= \bar{\bar{w}}\bar{\bar{y}}z + \bar{\bar{x}}\bar{\bar{z}} + \bar{\bar{w}}x$

$F = \bar{w}\bar{y}z + x\bar{z} + wx$

Fig. Logic diagram.

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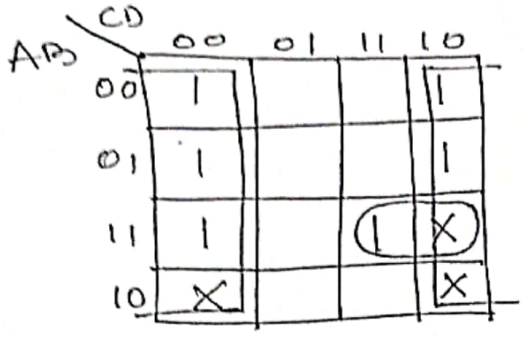
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Q. Simplify the following Boolean expression using K-Map & implement using NAND gates only.

$F(A,B,C,D) = \Sigma(0,2,4,6,12,15)$

$D(A,B,C,D) = \Sigma(8,10,14)$ [2019 Fall]

Sol^n:



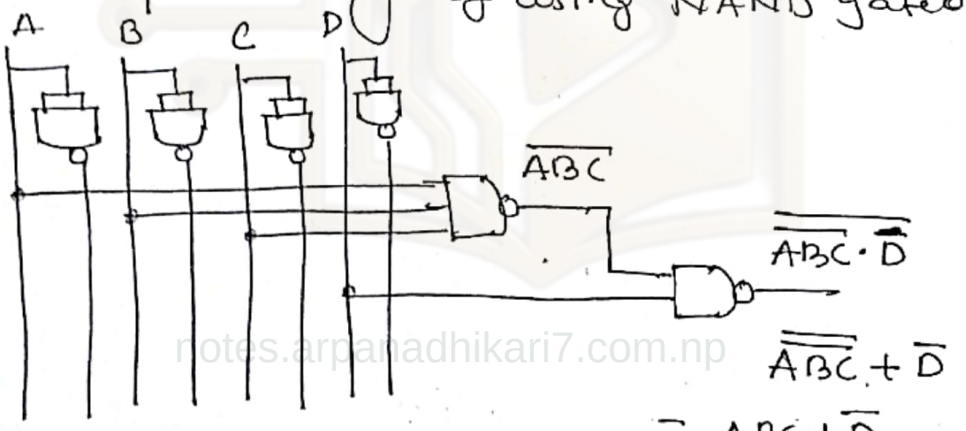
$F = ABC + \overline{D}$

Now, Double complementing on Both sides we get

$$\overline{\overline{F}} = \overline{\overline{ABC + \overline{D}}}$$

$$= \overline{\overline{ABC} \cdot \overline{D}}$$

Now, Implementing by using NAND gate only



$\overline{\overline{ABC} + \overline{D}}$

$= ABC + \overline{D}$

Fig. logic diagram.

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Q. Use K-Map to simplify the given function once by considering the don't care condition & once by ignoring the don't care condition & realize it using the basic gates
$F(A,B,C,D) = \Sigma(1,4,8,12,13,15)$ & don't care $d(A,B,C,D) = \Sigma(3,14)$ [2017 spring]

Sol:→ (i) First case
Considering the don't care condition
$F(A,B,C,D) = \Sigma(1,4,8,12,13,15)$
$d(A,B,C,D) = \Sigma(3,14)$

AB \ CD	00	01	11	10
00		1	X	
01	1			
11	1	1	1	X
10	1			

$F = \bar{A}\bar{B}D + B\bar{C}\bar{D} + A\bar{C}\bar{D} + AB$

Logic diagram

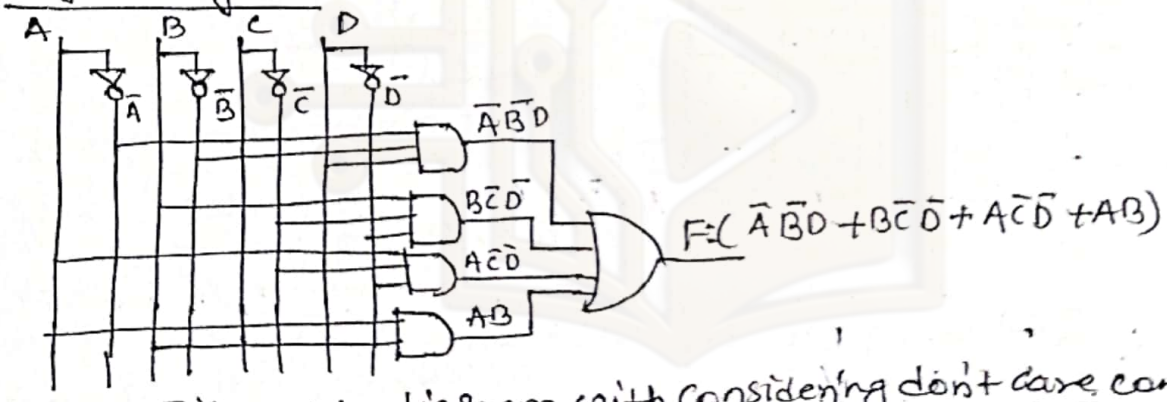


Fig. logic diagram with considering don't care condition

(ii) Second case.
Ignoring the don't care condition.
$F(A,B,C,D) = \Sigma(1,4,8,12,13,15)$

AB \ CD	00	01	11	10
00		1		
01	1			
11	1	1	1	
10	1			

$F = \bar{A}\bar{B}\bar{C}D + B\bar{C}\bar{D} + A\bar{C}\bar{D} + ABD$

) logic diagram in next page

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$F = \bar{A}\bar{B}\bar{C}D + B\bar{C}\bar{D} + A\bar{C}\bar{D} + ABD$

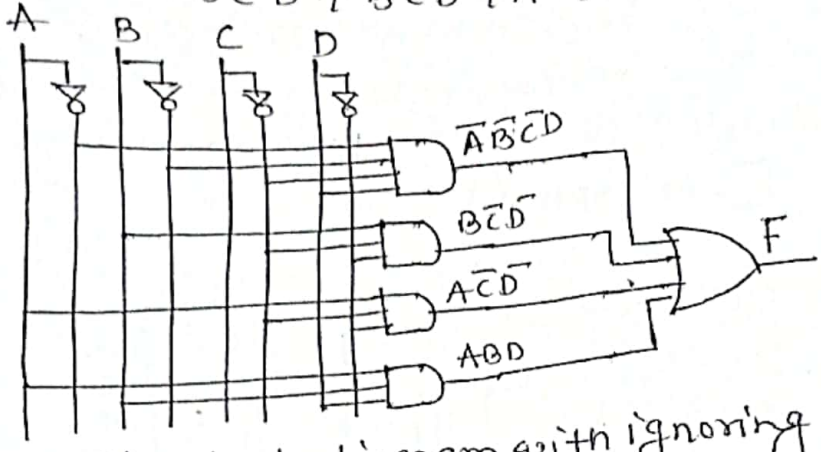
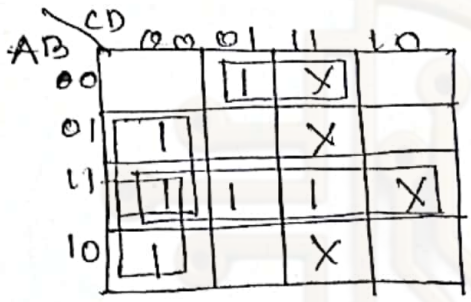


Fig:- logic diagram with ignoring the don't care cond

Q. Use K-Map to simplify the given Boolean function(s) don't care condition & realize it using basic gates only.
$F = \Sigma(1,4,8,12,13,15)$ & $d = \Sigma(3,7,11,14)$. (2018 Fall)

Sol:- Here Given.

$F = \Sigma(1,4,8,12,13,15)$ & $d = \Sigma(3,7,11,14)$



$F = \bar{A}\bar{B}D + B\bar{C}\bar{D} + A\bar{C}\bar{D} + AB$

Now, Realizing this expression using Basic gates only.

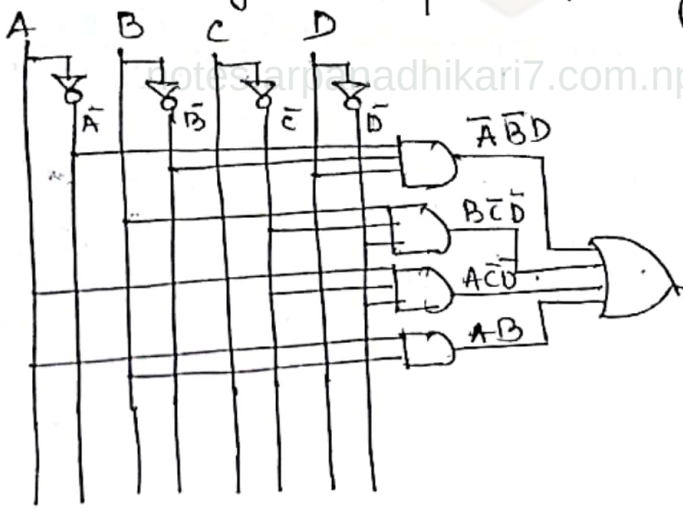


Fig. logic diagram.

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* Design a combinational circuit whose input is a four bit number and at the output, we obtain the two's complement of the input.

⇒ Let A_3, A_2, A_1, A_0 be the four bit input & B_3, B_2, B_1, B_0 represents the two's complement of input.

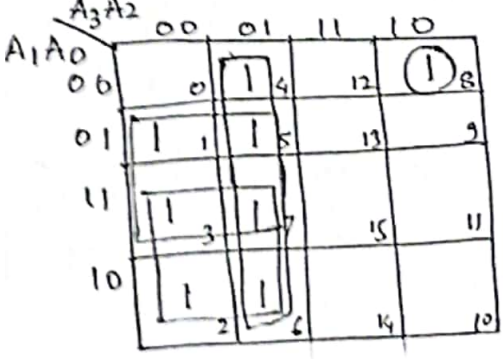
(i) First let us write the Truth Table.

Decimal	input				output			
	A_3	A_2	A_1	A_0	B_3	B_2	B_1	B_0
0	0	0	0	0	0	0	0	0
1	0	0	0	1	1	1	1	1
2	0	0	1	0	1	1	1	0
3	0	0	1	1	1	1	0	1
4	0	1	0	0	1	1	0	0
5	0	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	0
7	0	1	1	1	1	0	0	1
8	1	0	0	0	1	0	0	0
9	1	0	0	1	0	1	1	1
10	1	0	1	0	0	1	1	0
11	1	0	1	1	0	1	0	1
12	1	1	0	0	0	1	0	0
13	1	1	0	1	0	0	1	1
14	1	1	1	0	0	0	1	0
15	1	1	1	1	0	0	0	1

Truth Table.

K-Map Simplification.

For output B_3

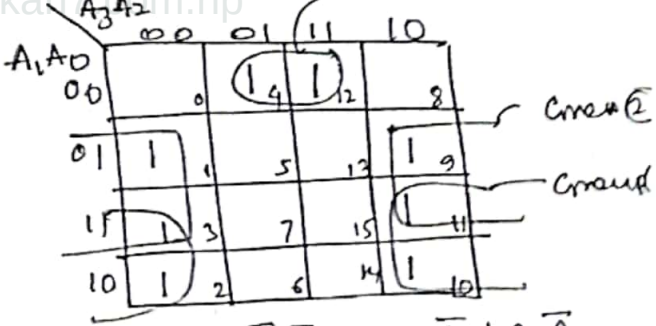


$B_3 = \overline{A_3}A_2 + \overline{A_1}\overline{A_0}A_3\overline{A_2} + A_0\overline{A_3}$

$= \overline{A_3}A_2 + \overline{A_1}\overline{A_0}A_3\overline{A_2} + A_1\overline{A_3}$

$= \overline{A_3}A_2 + \overline{A_1}\overline{A_0}A_3\overline{A_2} + \overline{A_3}(A_0+A_1)$

For output B_2 — Group ①



$B_2 = A_2\overline{A_1}\overline{A_0} + A_0\overline{A_2} + A_1\overline{A_2}$

$= \overline{A_0}\overline{A_1}A_2 + \overline{A_2}(A_0+A_1)$

But $\overline{A_0}\cdot\overline{A_1} = \overline{A_1+A_0}$ (De Morgan's)

$B_2 = A_2(\overline{A_0+A_1}) + \overline{A_2}(A_0+A_1)$

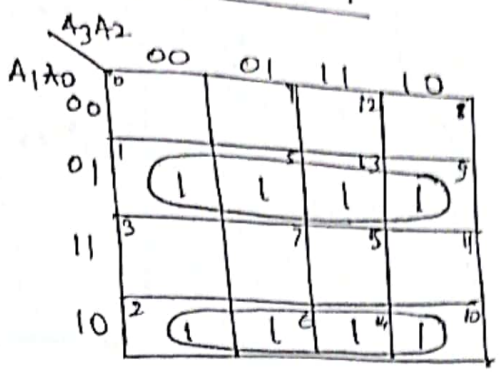
Let $A_0+A_1 = X$

$\therefore B_2 = A_2\overline{X} + \overline{A_2}X = A_2 \oplus X$

$B_2 = A_2 \oplus (A_0+A_1)$

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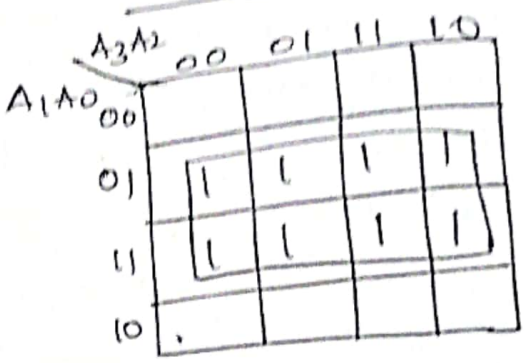
For output B_1



$B_1 = A_0\bar{A_1} + \bar{A_0}A_1$

$= A_0 \oplus A_1$

For output B_0

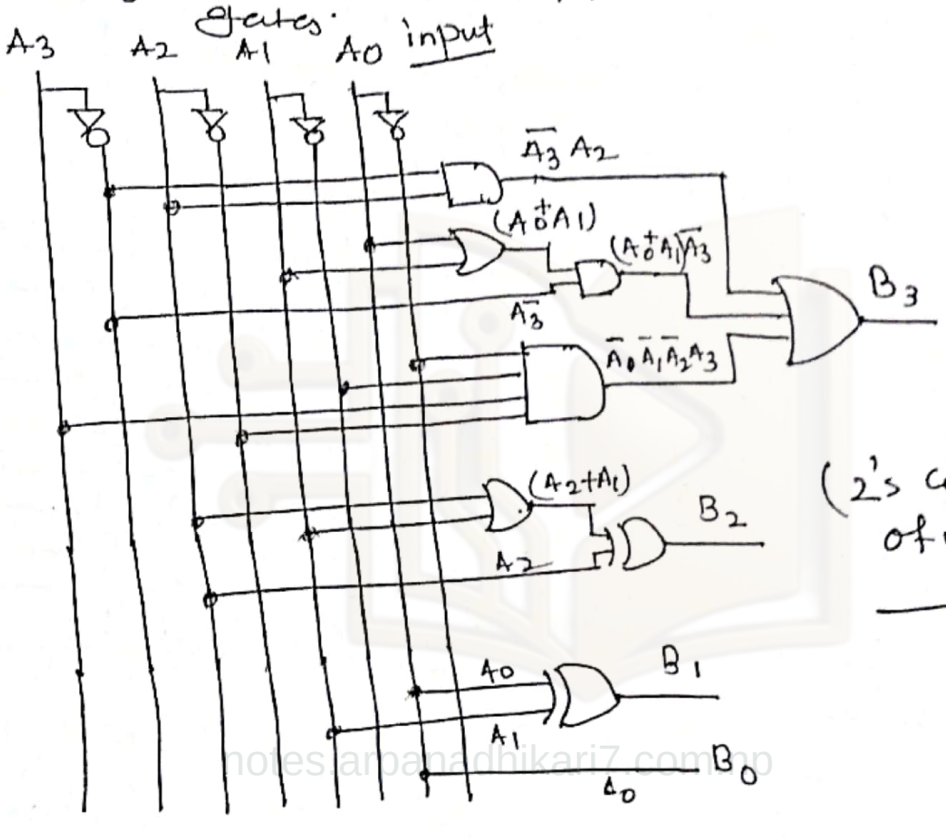


$B_0 = A_0$

$\therefore B_3 = \bar{A_3}A_2 + \bar{A_1}\bar{A_0}A_3\bar{A_2} + \bar{A_3}(A_0+A_1)$

$B_2 = A_2 \oplus (A_2+A_1)$

Finally let us realize the combinational ckt using ba gates.



(2's complement of input)

Fig: The Require Combinational CKt

Q. Design a code conversion ckt which converts BCD to excess-3 code (2012 fall) (Done in class)

Q. Design a combinational ckt that has four inputs & two One of the O/Ps is high when majority of i/ps are high Second O/p is high only when all inputs are of same typ (2017 spring) (Done in class)

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*Design a code conversion circuit which converts Binary code to Gray code. (2019 Fall)

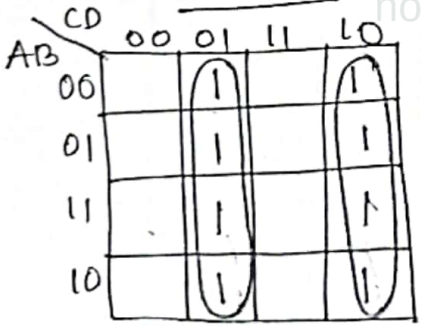
Sol:→ First let us write the truth table

Decimal	Binary code D C B A	Gray code G_3 G_2 G_1 G_0
0	0 0 0 0	0 0 0 0
1	0 0 0 1	0 0 0 1
2	0 0 1 0	0 0 1 1
3	0 0 1 1	0 0 1 0
4	0 1 0 0	0 1 1 0
5	0 1 0 1	0 1 1 1
6	0 1 1 0	0 1 0 1
7	0 1 1 1	0 1 0 0
8	1 0 0 0	1 1 0 0
9	1 0 0 1	1 1 0 1
10	1 0 1 0	1 1 1 1
11	1 0 1 1	1 1 1 0
12	1 1 0 0	1 0 1 0
13	1 1 0 1	1 0 1 1
14	1 1 1 0	1 0 0 1
15	1 1 1 1	1 0 0 0

Truth table of Binary code to Gray code.

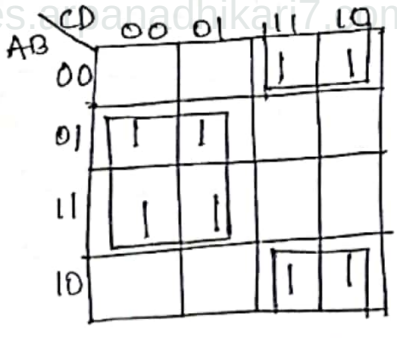
K-Map Simplification:

For G_0



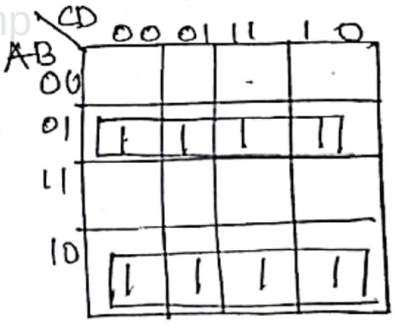
$G_0 = A\bar{B} + \bar{A}B$
$= A \oplus B$

For G_1



$G_1 = B\bar{C} + \bar{B}C$
$= B \oplus C$

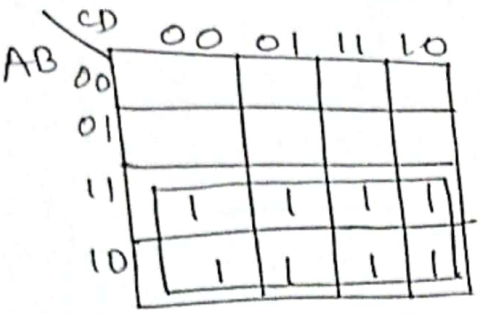
For G_2



$G_2 = C\bar{D} + \bar{C}D$
$= C \oplus D$

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For G_3



$G_3 = A$

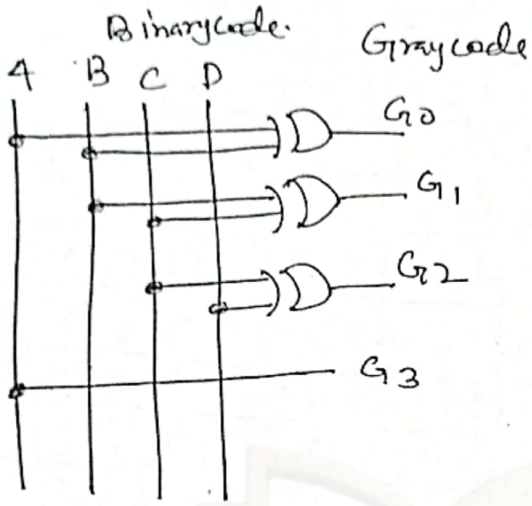


Fig: Logic diagram of Binary to gray code converter.

* Similarly we can design Gray code to Binary code converte

• Design a code converter circuit to convert BCD to gray co converter.

BCD code				Gray code			
B_3	B_2	B_1	B_0	G_3	G_2	G_1	G_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1

Table: Truth Table for BCD to gray code converter

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K-Map Simplification

For G_0

B_3B_2 \ B_1B_0	00	01	11	10
00		1		1
01		1		1
11	X	X	X	X
10		1	X	X

$G_0 = \overline{B_1} B_0 + B_1 \overline{B_0}$

$= B_1 \oplus B_0$

For G_1

B_3B_2 \ B_1B_0	00	01	11	10
00			1	1
01	1	1		
11	X	X	X	X
10			X	X

$G_1 = B_2 \overline{B_1} + \overline{B_2} B_1$

$= B_2 \oplus B_1$

For G_2

B_3B_2 \ B_1B_0	00	01	11	10
00				
01	1	1	1	1
11	X	X	X	X
10	1	1	X	X

$G_2 = B_2 + B_3$

For G_3

B_3B_2 \ B_1B_0	00	01	11	10
00				
01				
11	X	X	X	X
10	1	1	X	X

$G_3 = B_3$

Logic diagram

BCD code.

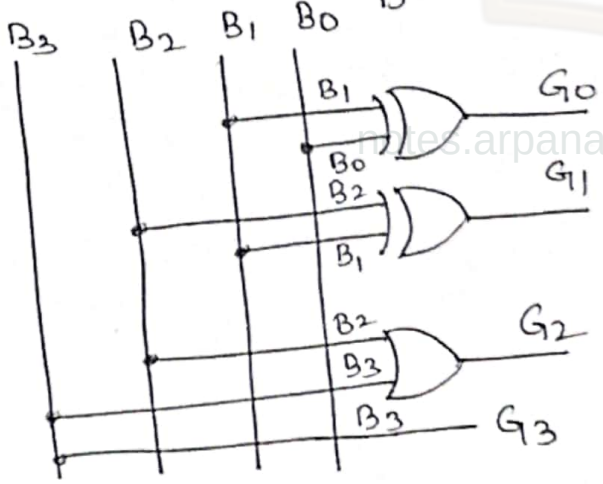


Fig: BCD to gray code converter

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(X) Design a combinational CKt that converts a decimal digit from the 2421 code to BCD (2016 Fall)

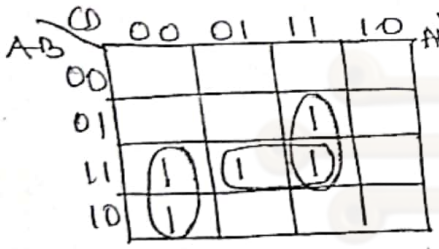
=) Let us write the truth table

Decimal Digit	2 4 2 1 D C B A	8 4 2 1 B_3 B_2 B_1 B_0
0	0 0 0 0	0 0 0 0
1	0 0 0 1	0 0 0 1
2	0 0 1 0	0 0 1 0
3	0 0 1 1	0 0 1 1
4	0 1 0 0	0 1 0 0
5	1 0 1 1	0 1 0 1
6	1 1 0 0	0 1 1 0
7	1 1 0 1	0 1 1 1
8	1 1 1 0	1 0 0 0
9	1 1 1 1	1 0 0 1

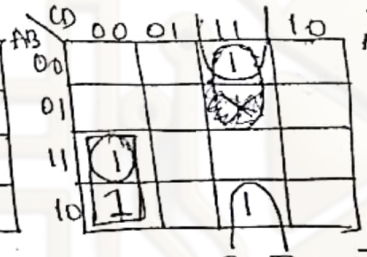
Truth Table of 2421 code to BCD code.

K-Map Simplification:

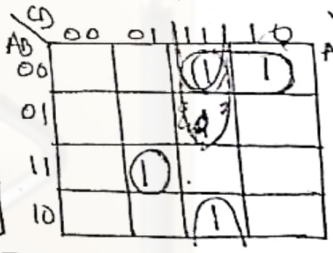
For output B_0



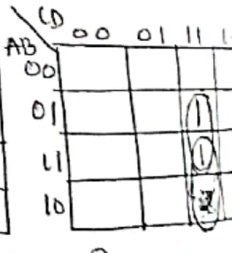
for B_1



for B_2



for B_3



$B_0 = A\bar{C}\bar{D} + BCD + A\bar{B}D$ $B_1 = A\bar{B}\bar{C}\bar{D} + \bar{A}CD + \bar{B}CD$ $B_2 = AB\bar{C}\bar{D} + \bar{A}\bar{B}C + \bar{B}CD$ $B_3 = BCD$

Finally let us realize the combinational CKt using basic gates

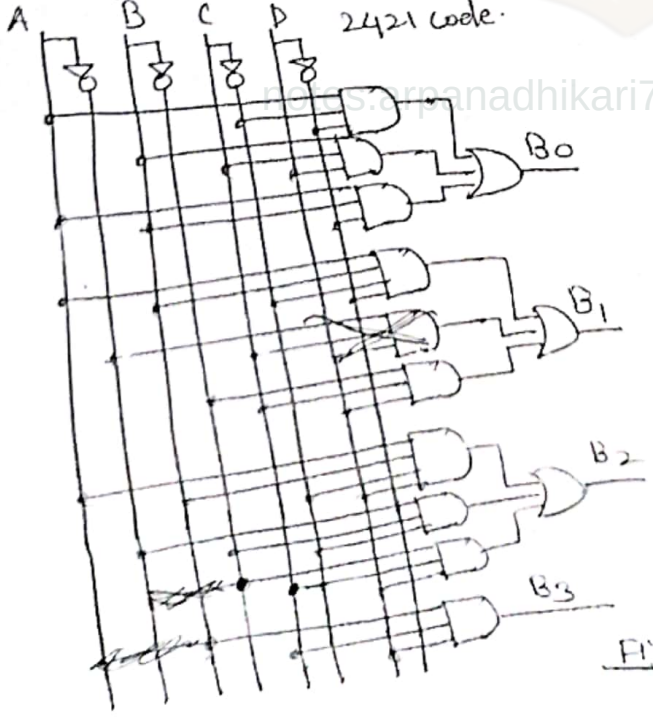


Fig: Required Combinational CKt for 2421 code to BCD

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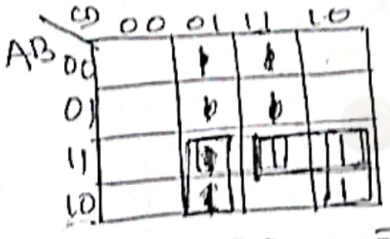
Q. Design a code converter ckt which converts 8 4-2-1 code to binary. [2012 spring]

Sol'n:→ First let us write the truth table.

Decimal	8 4 -2 -1 D C B A	Binary B_1 B_2 B_1 B_0
0	0 0 0 0	0 0 0 0
1	0 1 1 1	0 0 0 1
2	0 1 1 0	0 0 1 0
3	0 1 0 1	0 0 1 1
4	0 1 0 0	0 1 0 0
5	1 0 1 1	0 1 0 1
6	1 0 1 0	0 1 1 0
7	1 0 0 1	0 1 1 1
8	1 0 0 0	1 0 0 0
9	1 1 1 1	1 0 0 1

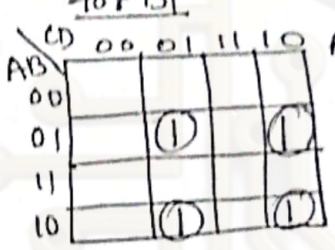
K-Map Simplification.

for B_0



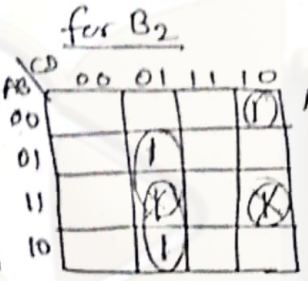
$B_0 = AB\bar{D} + AC\bar{D} + A\bar{C}D$

for B_1



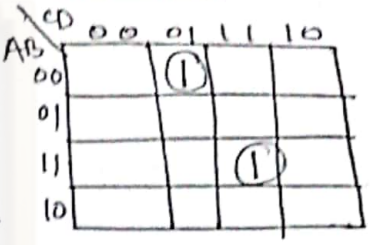
$B_1 = \bar{A}B\bar{C}D + \bar{A}BC\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D}$

for B_2



$B_2 = \bar{A}\bar{B}CD + A\bar{B}\bar{C}\bar{D} + A\bar{C}D$

for B_1



$B_1 = \bar{A}\bar{B}\bar{C}D + AB$

Finally, let us realize the combinational ckt using basic gat we set.

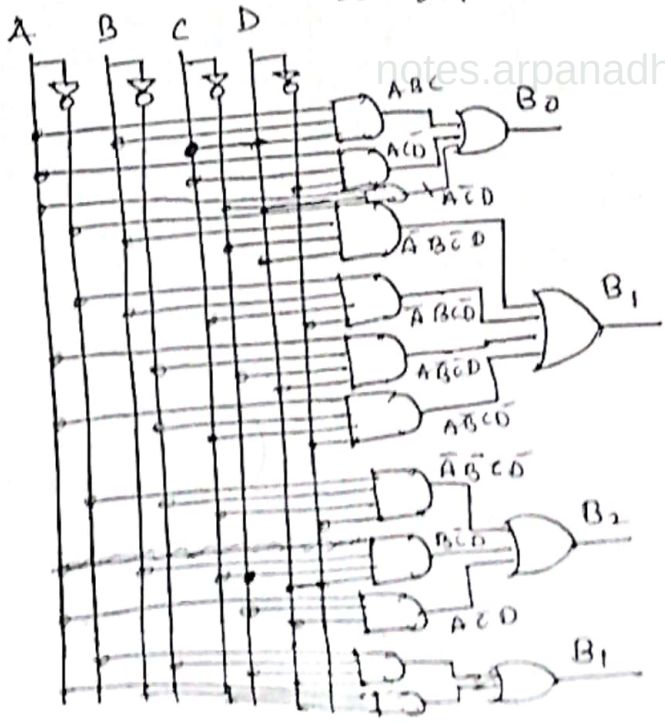


Fig Required ckt for 8-4-2-1 code to Binary.

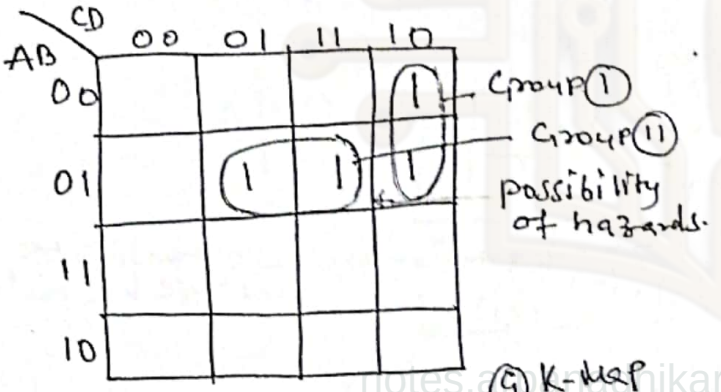
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* Design a combinational logic ckt that has ~~4~~ inputs Numbered 0 through to 9, and one output. The Output is required to go HIGH whenever any one or more of the input numbered 2, 5, 6, 7, 8, 9 is HIGH the ckt should be free of static hazards.

Sol^{n} ⇒ First let us write the Truth table.

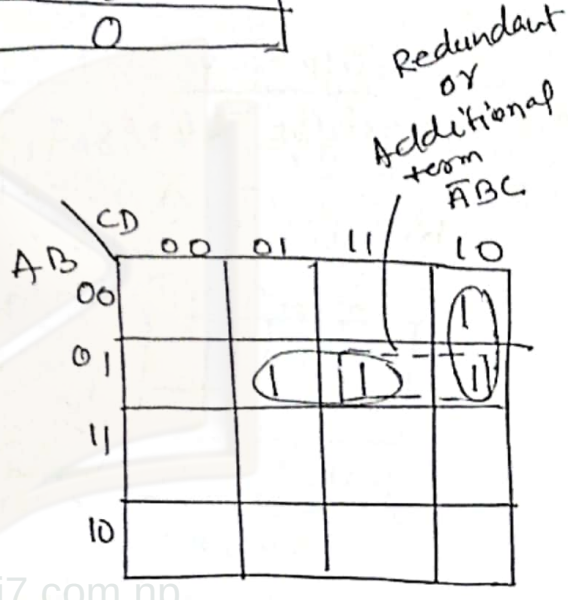
Inputs				Output Y
D	C	B	A	
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
:	:	:	:	:

K-Map simplification for output Y



(4) K-Map

$F = \bar{A}C\bar{D} + \bar{A}BD$



(5) Modified K-Map

In fig. the possible point where hazards can occur as shown above. With the normal grouping the expression of o/p will be given by

$F = \bar{A}C\bar{D} + \bar{A}BD$

from fig (b) the additional term $\bar{A}BC$ introducing in the ~~outp~~ expression

Therefore expression of the o/p of a hazard free ckt is given by.

$F = \bar{A}C\bar{D} + \bar{A}BD + \underbrace{\bar{A}BC}_{\text{New term}}$

The Hazard free ckt is given below

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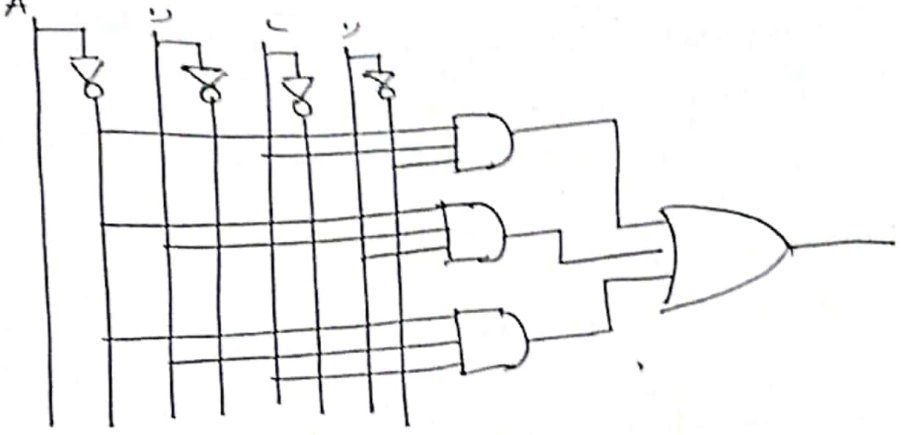
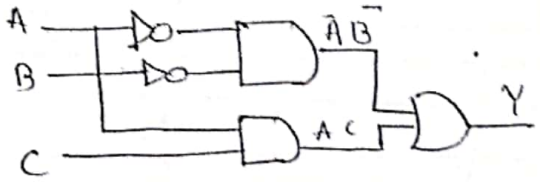


Fig. Realization or Required combinational ckt of Hazard free ckt

* Modify the ckt of fig. to make its hazard free



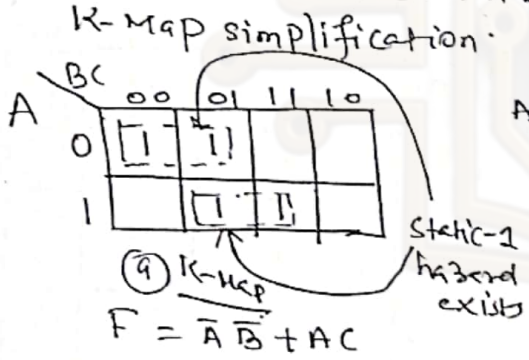
Soln

O/P of Y

$Y = \bar{A}\bar{B} + AC = \bar{A}\bar{B}(C+\bar{C}) + AC(B+\bar{B})$

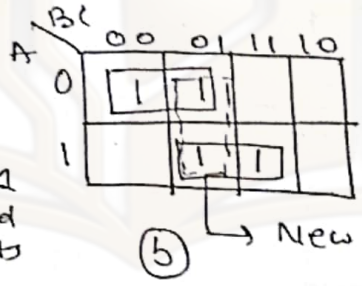
$= \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + ABC + A\bar{B}C$

K-Map simplification.



(a) K-Map

$F = \bar{A}\bar{B} + AC$



(b) New group $\bar{B}C$ to eliminate static 1-hazard

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The K-Map is Modified as shown in (b) to eliminate Static 1-ha

new expression for o/p will be.

$F = \bar{A}\bar{B} + AC + \underbrace{\bar{B}C}_{\text{Neo term.}}$

The modified ckt which is free of Static 1-hazards

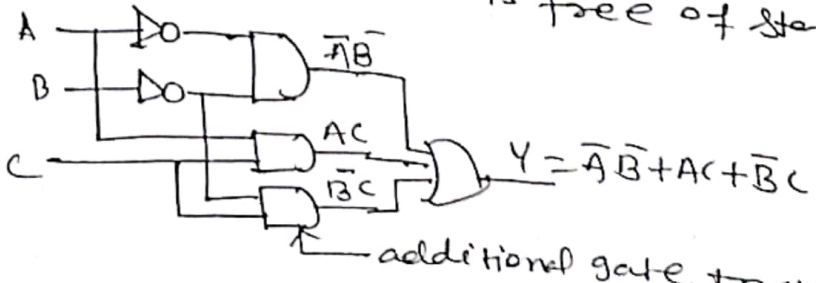


Fig. Hazard free ckt

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Q. Implement the following Boolean function $F = \Sigma(1,3,5$ using 4X1 MUX. (2014 Fall, 2014 Spring, 2015 Fall, Spring, 2016 Spring,

Sol'n Here, Given that

$F = \Sigma(1,3,5,6)$

Let us assume three Variable they are A, B, C.

$F(A,B,C) = \Sigma(1,3,5,6)$.

Now, Let us write the Truth table of a given functi

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Now, Next we write the design table or Implementation table.

	D_0	D_1	D_2	D_3	
$\bar{A}$	0	(1)	2	(3)	Row 1
A	4	(5)	(6)	7	Row 2
Input to MUX	0	1	A	$\bar{A}$	

Fig: Design table

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Lastly we draw the logic diagram.

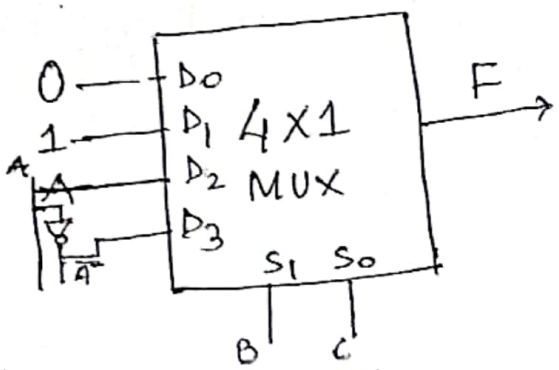


Fig. Logic diagram.

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* Construct 8X1 MUX using 4X1 MUX & explain with truth table. [2019 Fall]

Sol'n: The cascading of two 4X1 MUX results in 8X1 MUX is shown below.

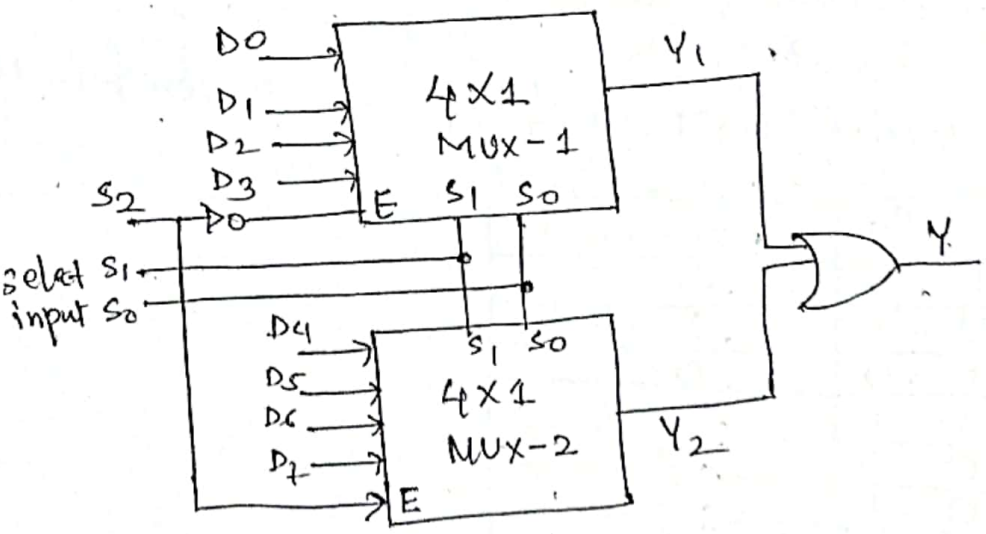


Fig: 8X1 MUX by cascading two 4X1 MUX

Truth Table

Select i/P			O/P
S_2	S_1	S_0	Y
0	0	0	D_0
0	0	1	D_1
0	1	0	D_2
0	1	1	D_3
1	0	0	D_4
1	0	1	D_5
1	1	0	D_6
1	1	1	D_7

MUX 1 is enable

MUX 2 is enable.

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There are in all eight data inputs (D_0 to D_7). The select lines S_1 & S_0 of both 4X1 MUX are connected in parallel whereas a third select i/P S_2 is used for enabling one MUX at a time. The output of the two Multiplexers are ORed to obtain the final o/p Y.

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4. Implement the following logic function using 4x1 MU

$F(A,B,C) = \Pi M(0,1,3,5,7)$

Sol^{n} → Here Given Function is

$F(A,B,C) = \Pi M(0,1,3,5,7)$

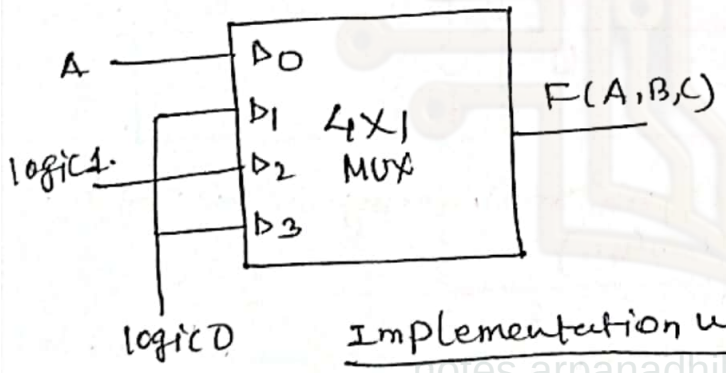
The given expression is in the POS form so, we must encir those maxterms which are not included in the given boole expression.

Implementation table

Inputs	D_0	D_1	D_2	D_3
$\bar{A}$	0	1	(2)	3
A	(4)	5	(6)	7
Inputs to MUX	A | 0	1	0	

encircle the maxterms which are not included in the given Boolean expressi

Logic diagram:



Implementation using 4X1 MUX

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5. Implement the following Boolean expression using 8X1M

$F(A,B,C,D) = \bar{A}\bar{B}\bar{D} + ABC + \bar{B}CD + \bar{A}CD$

Sol^{n} → (i) Let us convert the given expression in standard sop fo

$\bar{A}\bar{B}\bar{D}(C+\bar{C}) + ABC(D+\bar{D}) + \bar{B}CD(A+\bar{A}) + \bar{A}CD(B+\bar{B})$

$= \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + ABCD + ABC\bar{D} + A\bar{B}CD + \bar{A}\bar{B}CD + \bar{A}BCD + \bar{A}\bar{B}CD$

Now, $F(A,B,C,D) = \Sigma m(2,0,15,14,11,3,7,3)$

writing in arranging form

$F(A,B,C,D) = \Sigma m(0,2,3,7,11,14,15)$

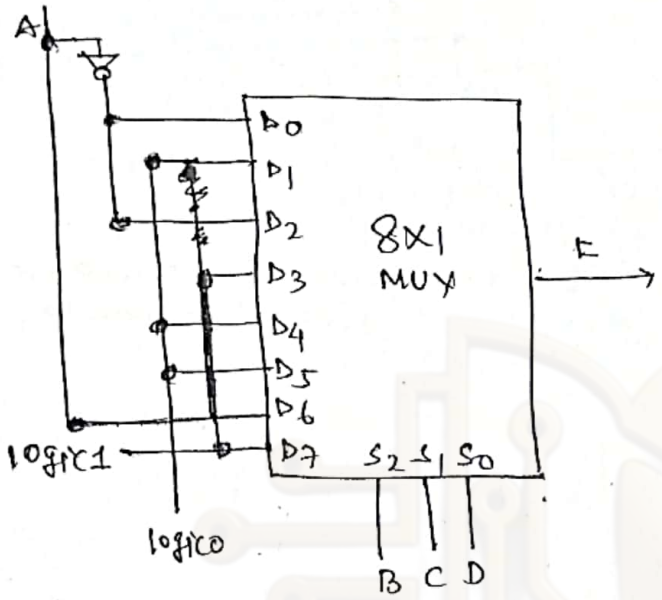
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Implementation table.

Inputs	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
$\bar{A}$	(0)	1	(2)	(3)	4	5	6	(7)
A	8	9	10	(11)	12	13	(14)	(15)
Input to MUX	$\bar{A}$	0	$\bar{A}$	1	0	0	A	1

(iii) logic diagram.



Implementation

* Implement the following expression using suitable Multiplexer.

$F(A,B,C) = \sum m(0,2,4,6)$

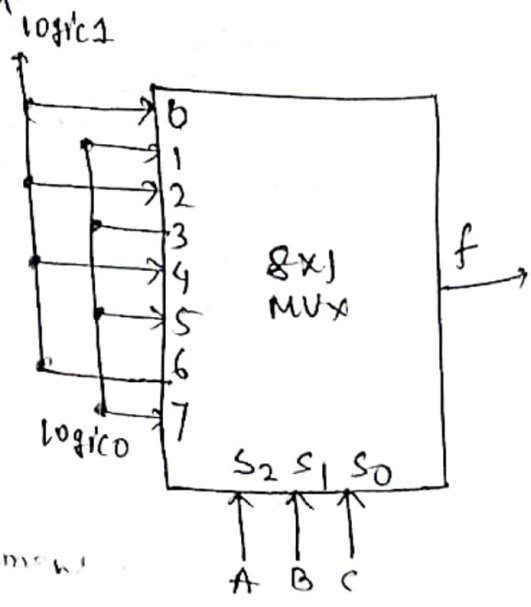
Sol^n $F(A,B,C) = \sum m(0,2,4,6)$

Here, is three variable, MUX having three select i/p must be used.

Therefore 8X1 MUX must be used.

Implementation table

D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
0	1	2	3				



Implementation

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4. Implement the following [illegible] $Y(A,B,C,D) = \Sigma(0,1,2,5,9,11,13,15)$. (2012 Fall,

Sol^n

Here, Given, $Y = (A,B,C,D) = \Sigma(0,1,2,5,9,11,13,15)$

First, Let us write the truth table of given function

A	B	C	D	Y
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

(i)

Next we write the Implementation table.

	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
A'	(0)	(1)	(2)	3	4	(5)	6	7
A	8	(9)	10	(11)	12	(13)	14	(15)
Input to MUX	A'	1	A'	A	0	1	0	A

Table: Implementation table

Lastly we draw the logic diagram

Next page. PTO

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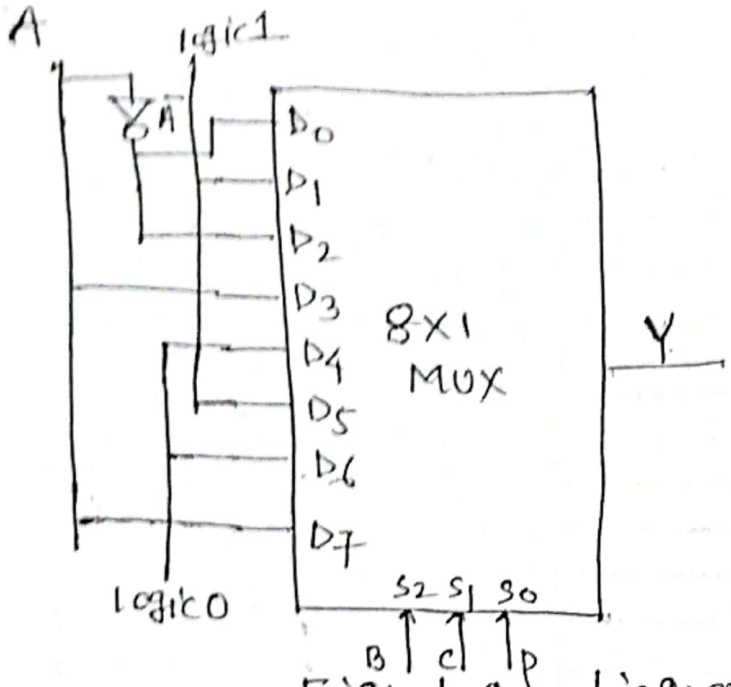


Fig: logic diagram.

Q. Implement the following with appropriate MUX.

(ii) $F(A,B,C,D) = \Sigma(0,1,3,4,8,9,15)$ (2014 Fall, 2015 Spring)

Sol^{n} Here Given. $(A,B,C,D) = \Sigma(0,1,3,4,8,9,15)$

Here is Four Variable, MUX having four select line. So use 16×1 MUX.

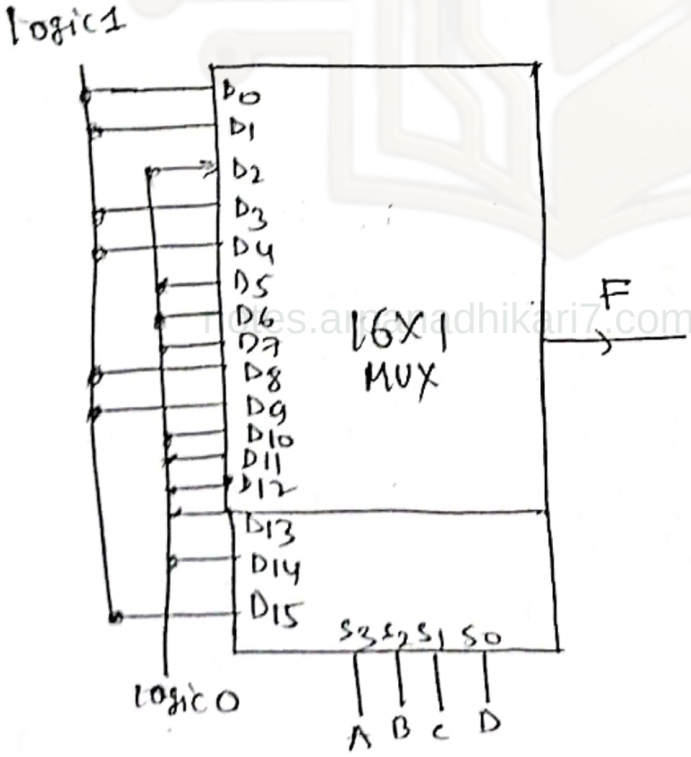


Fig: logic diagram of given function with appropriate MUX (16×1).

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Q. Implement the Full Subtractor using a 1X8 DEMUX

Sol^{n}:→ The Full Subtractor has three inputs A, B and B_{in} & two O/Ps namely Difference D & borrow out (B_{out}). The Truth Table of Full Subtractor is

Inputs			Outputs	
A	B	Bin	D	Bout
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Table:- Truth Table of a F.S

we express the difference & borrow out Outputs in Standard SOP form as

$D = \sum m(1,2,4,7)$

$B_{out} = \sum m(1,2,3,7)$

we connect D_{in} to logic 1 permanently & connect A, B & Bin to select S_2, S_1, S_0 respectively.

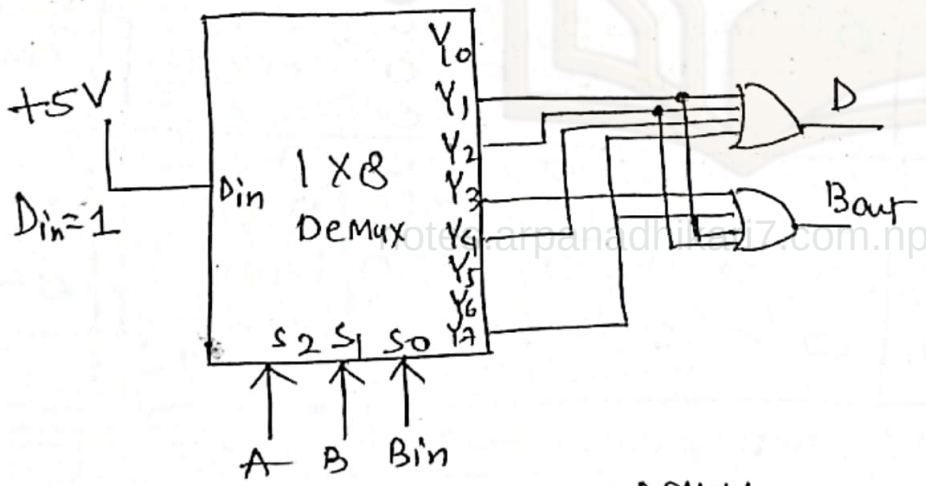


Fig:- F.S using 1X8 DEMUX

Q. Implement a Full adder using Demultiplexer

Sol^{n}:→ Similarly, we can do as Full Subtractor.

(DIY)

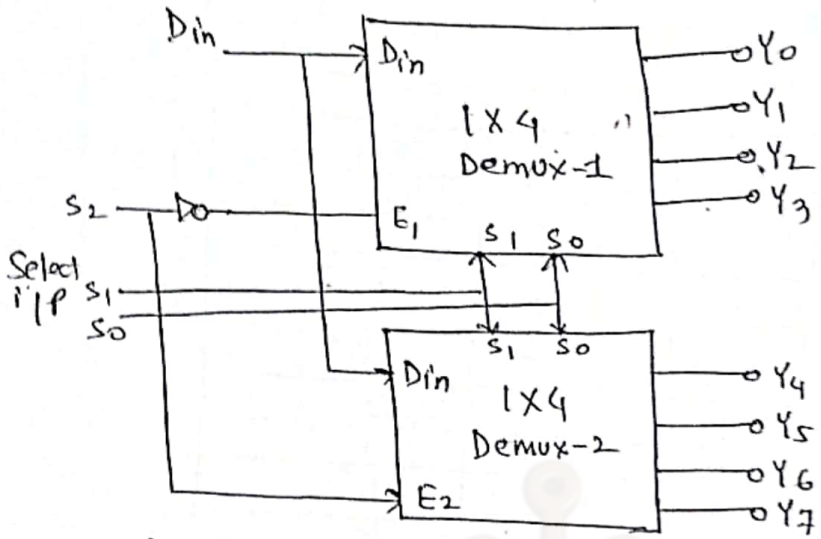
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* Obtain a 1X8 line DEMUX using two 1X4 line DEMUX.

Sol'n →



1X8 DEMUX using two 1X4 DEMUX

The select line S_1 & S_0 of the two 1X4 DEMUX are connected Parallel with each other & S_2 is used for selecting of t two 1X4 DEMUX. S_2 is connected directly to enable (E) i/p DEMUX-2 whereas inverted S_2 is connected to the enable i/ of DEMUX-1. The T.T of this ckt is shown below.

S_2	S_1	S_0	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7	
0	0	0	Din	0	0	0	0	0	0	0	DeMux enabl
0	0	1	0	Din	0	0	0	0	0	0	
0	1	0	0	0	Din	0	0	0	0	0	
0	1	1	0	0	0	Din	0	0	0	0	
1	0	0	0	0	0	0	Din	0	0	0	DeMux enable
1	0	1	0	0	0	0	0	Din	0	0	
1	1	0	0	0	0	0	0	0	Din	0	
1	1	1	0	0	0	0	0	0	0	Din	

Table: T.T of 1X8 DeMux using two 1X4 DeMux

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Q. Implement a Full adder using 4x1 MUX.

Soln:→ (i) First let us write the truth table of Full adder

inputs			outputs	
A	B	Cin	Sum(S)	Carry(C)
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

the sum & carry outputs can be expressed in the standard SOP form

$S = \Sigma m(1,2,4,7)$ & $C = \Sigma m(3,5,6,7)$

(ii) Now we write the Implementation table for Sum & Carry Output.

For sum

	D0	D1	D2	D3
$\bar{A}$	0	(1)	(2)	3
A	(4)	5	6	(7)
Input to MUX	A	$\bar{A}$	$\bar{A}$	A

For carry

	D0	D1	D2	D3
$\bar{A}$	0	1	2	(3)
A	4	(5)	(6)	(7)
Input to MUX	0	A	A	1

Table:- Implementation table for sum & carry.

(iii) Logic diagram

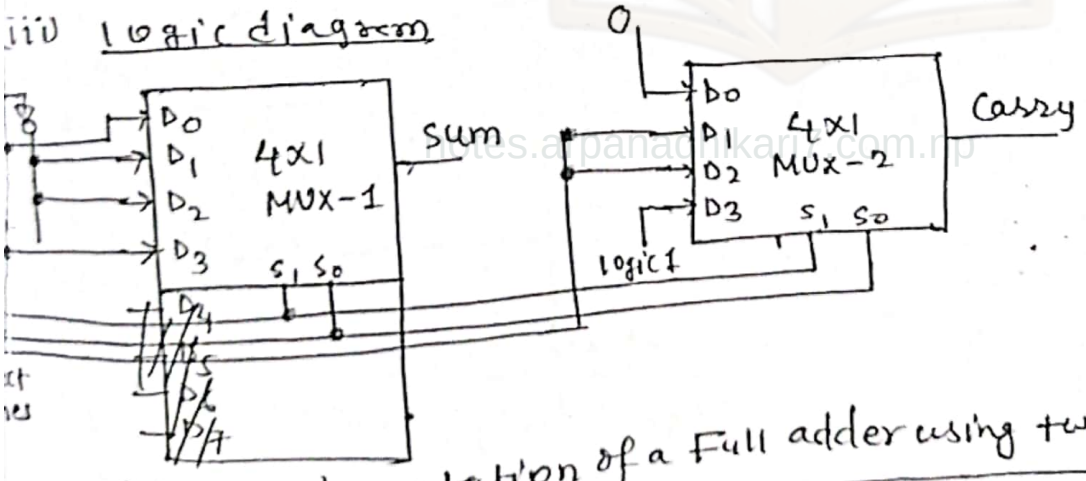


Fig:- Implementation of a Full adder using two 4x1 MUX.

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Q. Implement a Full adder using 8X1 MUX.

Soln: (i) First let us write the truth table of a F.A

Inputs			outputs	
A	B	Cin	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

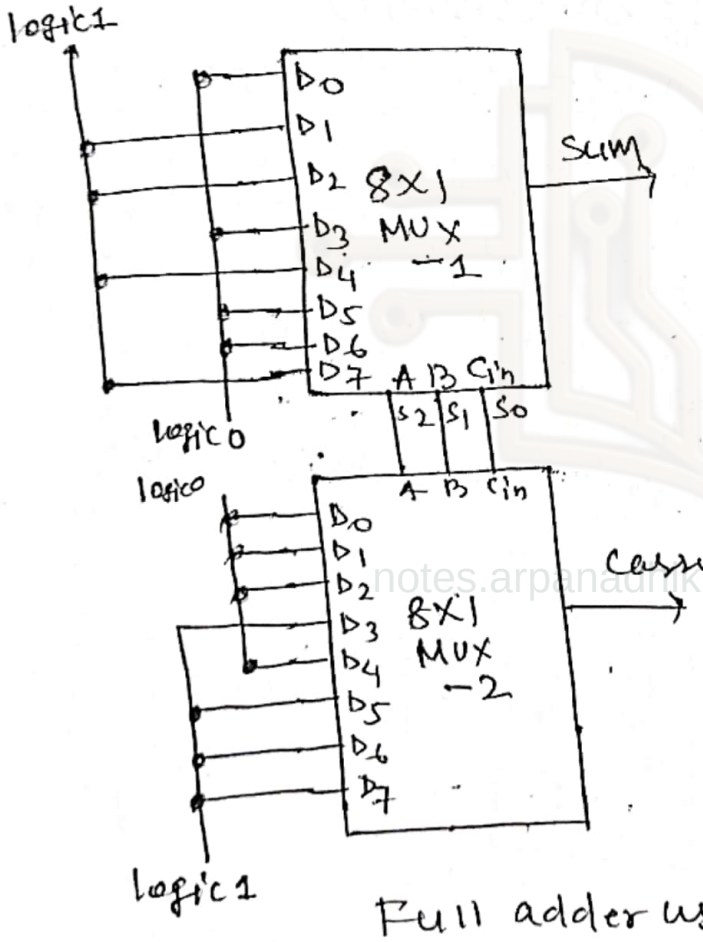
(ii) The sum and carry can be expressed in the SOP form

$$S = \sum m(1,2,4,7) \quad \& \quad C = \sum m(3,5,6,7)$$

(iii) We realize the ckt using two 8X1 MUX.

Remem

8 inputs th
using
$2^n = 2^3$
$n = 3$ sel



Full adder using 8X1 MUX

Note: (we can also realize ckt separately for sum & carry)

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Implement 4 to 1 multiplexer ... of ...

Ans: Implementation is shown below

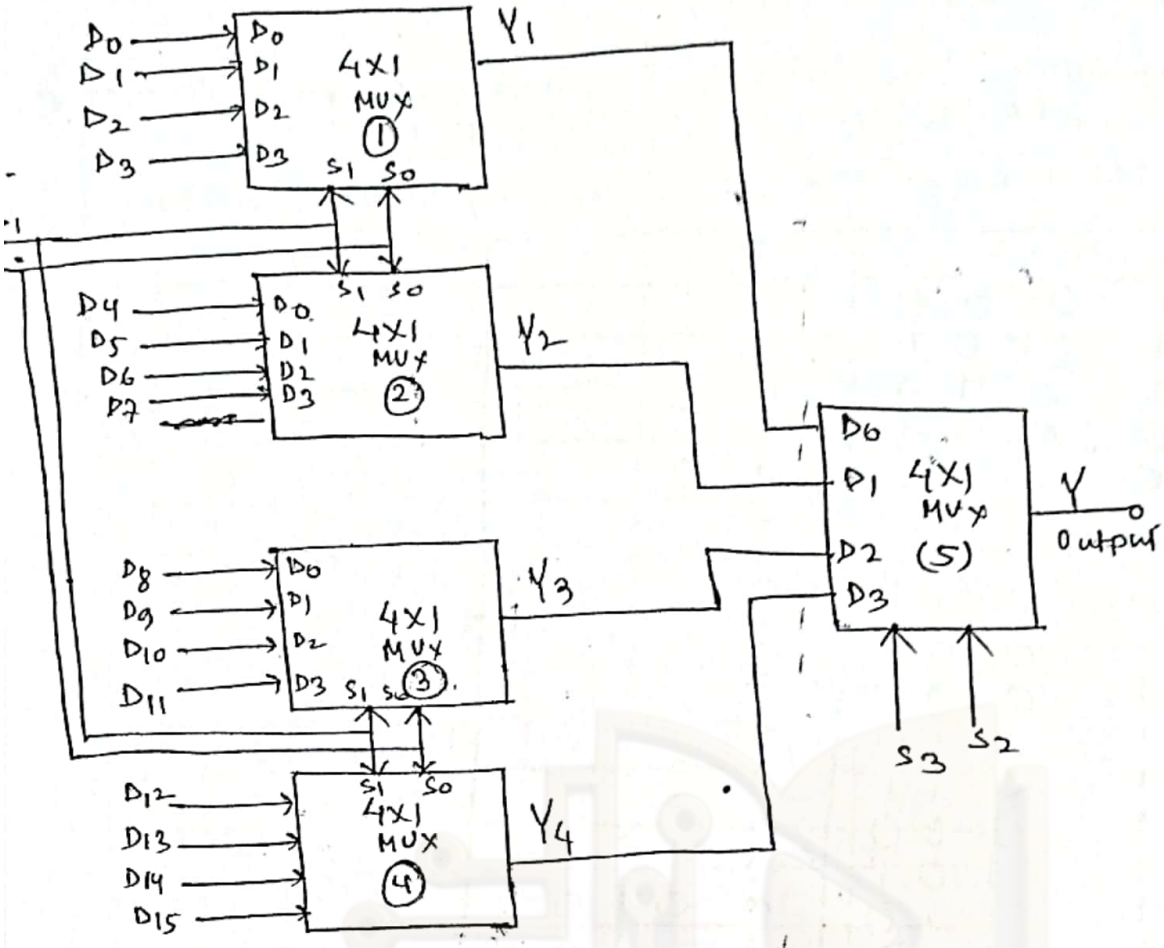


Fig: 16X1 MUX using 4X1 MUX

Function table of above ckt

				MUX O/P				Final O/P Y	
S_3	S_2	S_1	S_0	Y_1	Y_2	Y_3	Y_4		
0	0	0	0	D_0	D_4	D_8	D_{12}	D_0	$S_3S_2=0$ Therefore MUX-5 selects Y
0	0	0	1	D_1	D_5	D_9	D_{13}	D_1	
0	0	1	0	D_2	D_6	D_{10}	D_{14}	D_2	
0	0	1	1	D_3	D_7	D_{11}	D_{15}	D_3	
0	1	0	0	D_0	D_4	D_8	D_{12}	D_4	$S_3S_2=01$ MUX-5 select-
0	1	0	1	D_1	D_5	D_9	D_{13}	D_5	
0	1	1	0	D_2	D_6	D_{10}	D_{14}	D_6	
0	1	1	1	D_3	D_7	D_{11}	D_{15}	D_7	
1	0	0	0	D_0	D_4	D_8	D_{12}	D_8	$S_3S_2=1$ Therefore MUX- selec
1	0	0	1	D_1	D_5	D_9	D_{13}	D_9	
1	0	1	0	D_2	D_6	D_{10}	D_{14}	D_{10}	
1	0	1	1	D_3	D_7	D_{11}	D_{15}	D_{11}	
1	1	0	0	D_0	D_4	D_8	D_{12}	D_{12}	$S_3S_2=$ Therel MUX- selec
1	1	0	1	D_1	D_5	D_9	D_{13}	D_{13}	
1	1	1	0	D_2	D_6	D_{10}	D_{14}	D_{14}	
1	1	1	1	D_3	D_7	D_{11}	D_{15}	D_{15}	

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The select input S_1 & S_0 of MUX 1, 2, 3 & 4 are connected together. The select i/p S_3 & S_2 are applied to MUX-5. The O/p Y_1, Y_2, Y_3, Y_4 are applied to data i/p D_0, D_1, D_2 & D_3 of MUX-5 as shown in fig.

Q. Implement the following function with multiplexer.

Sol: $F(A,B,C,D) = \Sigma(0,1,3,4,8,9,15)$

First let us write the Truth table of given Function.

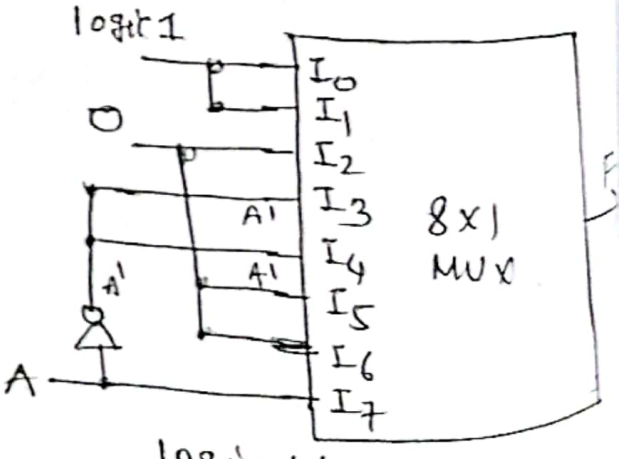
A	B	C	D	F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

Table → T.T

Implementation Table.

i/p	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
A'	(0)	(1)	2	(3)	(4)	5	6	7
A	(8)	(9)	10	11	12	13	14	(15)
4 to 1 MUX	1	1	0	A'	A'	0	0	A

Finally draw the logic diagram



logic diagram

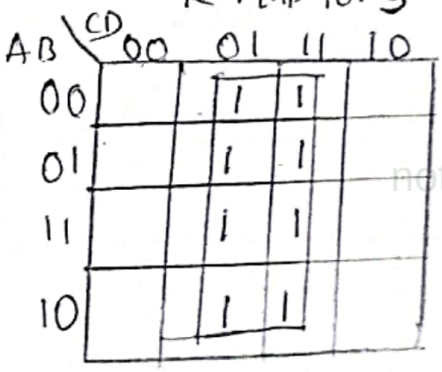
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8. Design a combinational circuit to produce 1's complement of a 4-bit binary number.

Soln:→

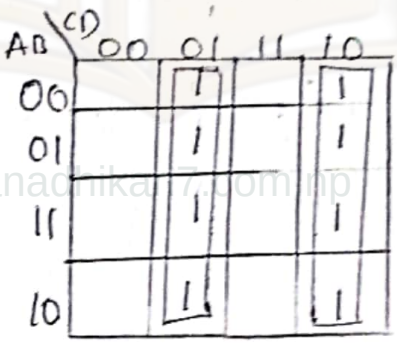
Decimal No.	A	B	C	D	P	Q	R	S
0	0	0	0	0	0	0	0	0
1	0	0	0	1	1	1	1	1
2	0	0	1	0	1	1	1	0
3	0	0	1	1	1	1	0	1
4	0	1	0	0	1	1	0	0
5	0	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	0
7	0	1	1	1	1	0	0	1
8	1	0	0	0	1	0	0	0
9	1	0	0	1	0	1	1	1
10	1	0	1	0	0	1	1	0
11	1	0	1	1	0	1	0	1
12	1	1	0	0	0	1	0	0
13	1	1	0	1	0	0	1	1
14	1	1	1	0	0	0	1	0
15	1	1	1	1	0	0	0	1

K Map for S



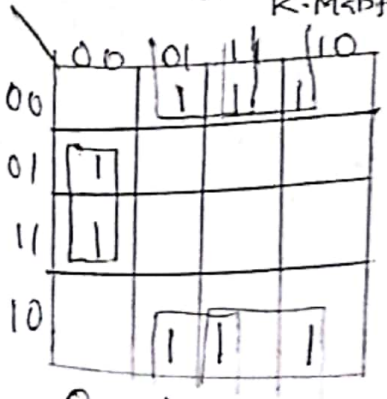
$S = D$

K Map for R



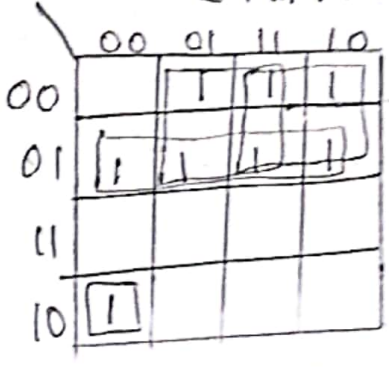
$R = \bar{C}D + C\bar{D} = C \oplus D$

K-Map for Q



$Q = \bar{B}C + \bar{B}D + B\bar{C}\bar{D}$

K-Map for P

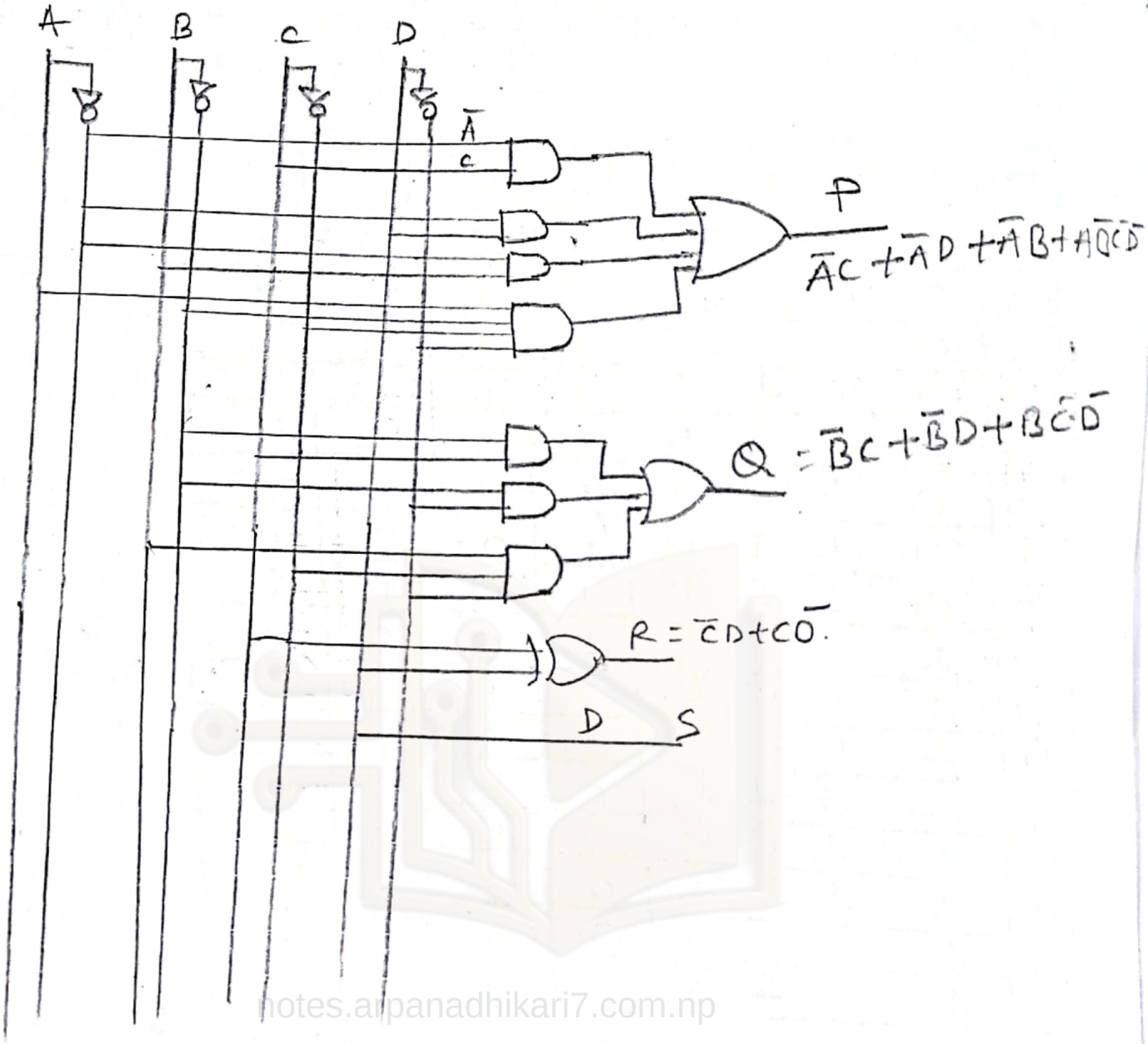


$P = \bar{A}C + \bar{A}D + \bar{A}B + A\bar{B}\bar{C}\bar{D}$

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Q. Design 3-bit Synchronous UP counter by using JK F/F

Sol^{n} →

3-Bit = 3 F/F

No of state $N = 2^n$

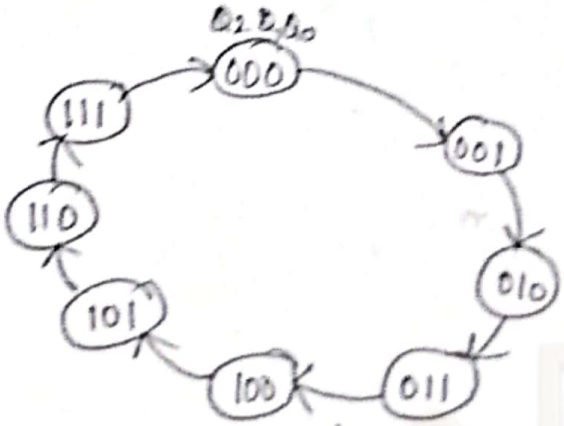
$n = 3$ $N = 2^3 = 8$ state

Max count $2^n - 1 = 2^3 - 1 = 8 - 1 = 7$

Count 0 to 07

Excitation Table of JK F/F

Q(t)	Q(t+1)	JK
0	0	0 X
0	1	1 X
1	0	X 1
1	1	X 0



State diagram

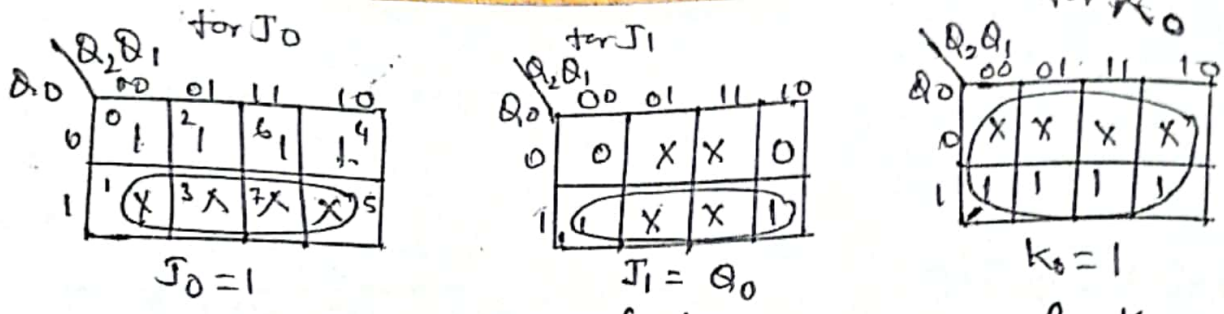
CKt Excitation Table

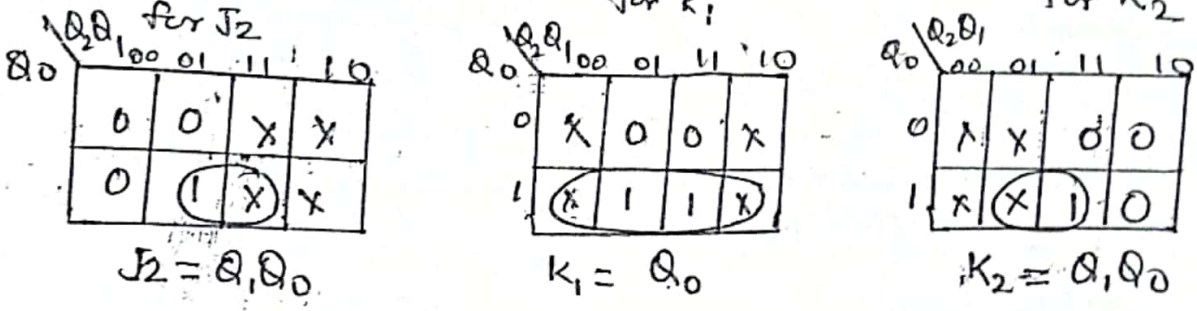
Present state $Q_2\ Q_1\ Q_0$	Next state $Q_2\ Q_1\ Q_0$	$J_0\ K_0$	$J_1\ K_1$	$J_2\ K_2$
0 0 0	0 0 1	1 X	0 X	0 X
0 0 1	0 1 0	X 1	1 X	0 X
0 1 0	0 1 1	1 X	X 0	0 X
0 1 1	1 0 0	X 1	X 1	1 X
1 0 0	1 0 1	1 X	0 X	X 0
1 0 1	1 1 0	X 1	1 X	X 0
1 1 0	1 1 1	1 X	X 0	X 0
1 1 1	0 0 0	X 1	X 1	X 1

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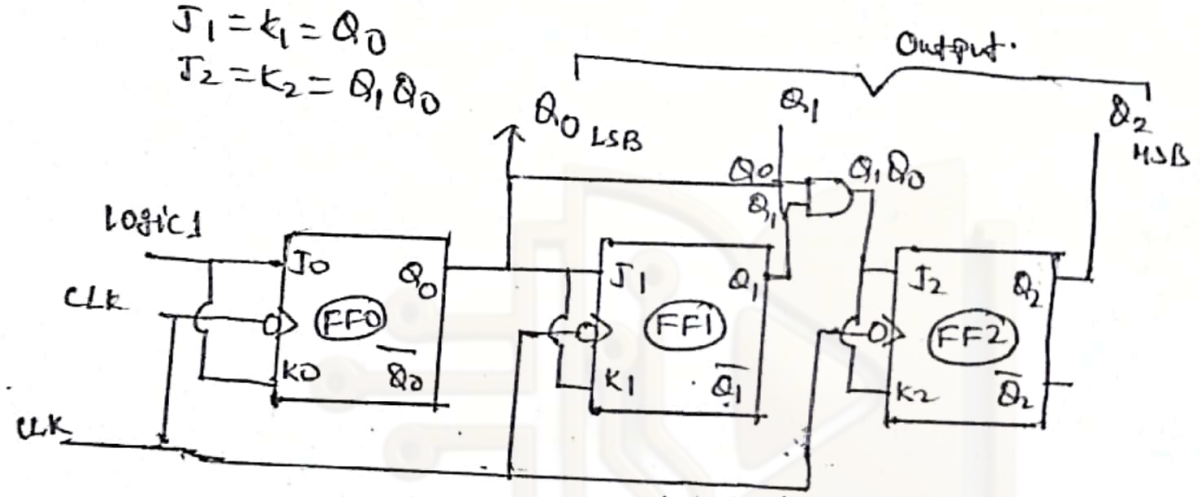




$J_0 = K_0 = 1$

$J_1 = K_1 = Q_0$

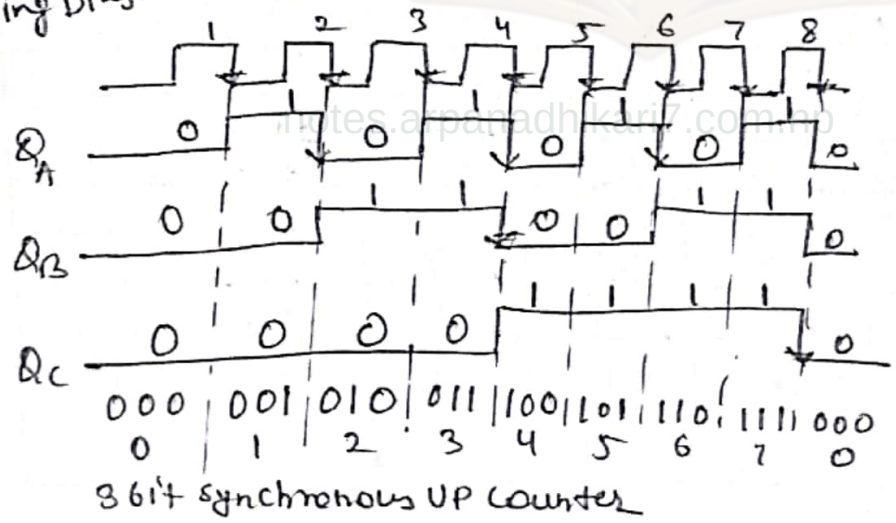
$J_2 = K_2 = Q_1Q_0$



Logic diagram.

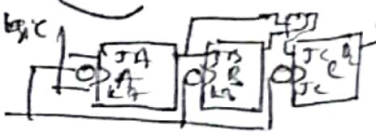
Note:

Timing Diagram.



3 bit synchronous UP counter

CLK	Q_C	Q_B	Q_A
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1



Here, Q_A changes on each CP as we progress from its original state to its final state & then back to its original state. To produce this operation F/F A is held in the toggle mode by connecting J & K i/p to High. for F/F B toggle when Q_A is 1. when Q_A is 0 F/F B is in no-change mode. remains its F/F C has to change only when Q_B & Q_A are at 1.

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Q. Design 2-bit synchronous UP counter using JK F/F.

Sol:

2-bit = 2 FlipFlop

No. of states $N = 2^n$

$n = 2 \quad N = 2^2 = 4$ state

Max. count $2^n - 1 = 2^2 - 1 = 4 - 1 = 3$

Count = 0 to 3

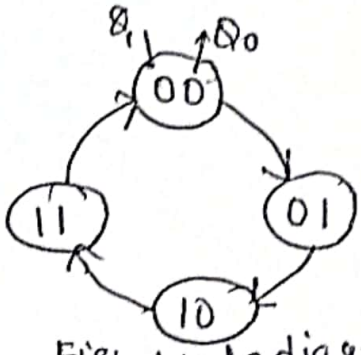


Fig: state diagram

Excitation Table of JK F/F

Q(t)	Q(t+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

CKT Excitation Table.

PS		NS		I/P			
Q_1	Q_0	Q_1	Q_0	J_1	K_1	J_0	K_0
0	0	0	1	0	X	1	X
0	1	1	0	1	X	X	1
1	0	1	1	X	0	1	X
1	1	0	0	X	1	X	1

for J_1

Q_1 \ Q_0	0	1
0	0	1
1	X	X

$J_1 = Q_0$

for K_1

Q_1 \ Q_0	0	1
0	X	X
1	0	1

$K_1 = Q_0$

for J_0

Q_1 \ Q_0	0	1
0	1	X
1	1	X

$J_0 = 1$

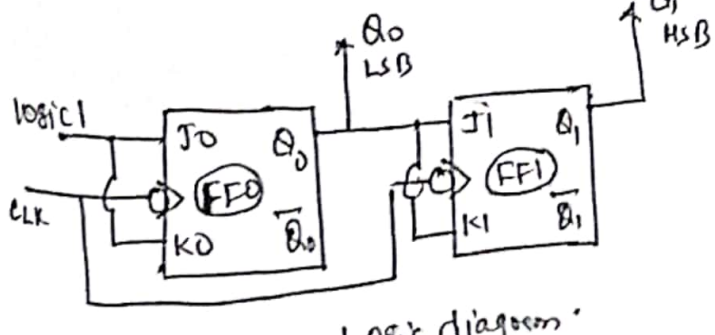
for K_0

Q_1 \ Q_0	0	1
0	X	1
1	X	1

$K_0 = 1$

$J_0 = K_0 = 1$

$J_1 = K_1 = Q_0$



Logic diagram.

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